

Final Year Design Project

Beauty Blush Salon



by

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For

Beauty Blush Salon

Under the Supervision of

Hasan Raza

Session: 2022-2026

Project ID: 22-KS-BSIT-10

Faculty of Computing & Information Technology,

University of the Punjab, Lahore, Pakistan.

(2022-2026)

DECLARATION

The work reported in this project is carried by Sahrish Yaseen (Roll no. 089351) and Mahnoor (Roll no. 089286), under the supervision of Project Supervisor Hasan Raza, at Govt. Graduate College, Civil Lines, Sheikhpura. We hereby declare that this project and the contents of the project are the product of our research and no part has been copied from copyrighted written or published sources. (Accept the references, standard mathematical or genetics models/equation/formulas/protocol, etc.). We further declare that this work has not been submitted for the award of any other degree diploma.

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STATEMENT OF SUBMISSION

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It is certified that this Sahrish Yaseen (089351) and Mahnoor (089286) have prepared documentation for their final year project (FYP) titled Beauty Blush Salon at the Department of Computer Science, Government Graduate College, Civil Lines, Sheikhpura, in the partial fulfillment of the requirements for the degree of Bachelor of Science in Information Technology.

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Acknowledgement

we would like to express our sincere thanks to Sir Hasan Raza, Lecturer at Government Graduate College, Civil, Sheikhpura, for his valuable guidance, encouragement and support during the preparation of our project documentation.

We are also thankful to our friends and families for their quiet support and motivation, which helped us stay focused throughout this phrase of our work.

Date: January 28, 2025

PREFACE

This is a final year project (FYP) report written by students of bachelor of Science in Information Technology at the University of the Punjab. We are Sahrish Yaseen and Mahnoor, both of us are studying in BSIT at Government Graduate College, Civil Lines, Sheikhupura.

To enable you to understand the idea of our project and the implementation process in detail and systematically, this report is divided into topics; each topic with a specific focus. We hope this approach allows you to read and understand our project report easily. Lastly, if you have any comments, you are welcome to contact us at:

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Executive Summary

The Beauty Blush Salon project will aim to develop a web-based salon appointment booking system. At Beauty Blush Salon, appointments are currently managed manually. This manual process causes issues such as time clashes, record mismanagement and time consumption. This project will convert the manual booking process into a digital and organized system. The system will allow clients to create accounts and log in. clients will be able to view available services, check prices, select their preferred stylist and book appointments according to available time slots. The system will support partial online payment to confirm appointments. Each client will have personal dashboard to view upcoming appointments and booking history. An admin panel will be provided where the admin will manage services, stylists time slots and appointments. All stylist availability will be controlled by the admin. This will help reduce manual work, save time and prevent scheduling conflicts. The project Proposal will explain the problem, objectives and scope of the system. The Software Requirements Specification (SRS) will define all functional and non-functional requirements along with use case. The Software Design Specification (SDS) will describe the system design, database structure and main modules. A prototype will visually demonstrate key features such as login, service selection, appointment booking and payment confirmation. Overall, this project will provide a simple, modern and efficient solution for managing salon appointments.

FYDP Overview

FYDP Title _ Beauty Blush Salon

Sr. No	Roll Numbers	Name	Signatures
1.	089351	Sahrish Yaseen	
2.	089286	Mahnoor	
3.			
4.			
5.			

Table 1 Project Proposal Summary

FYDP Goals
Clients will be able to book and cancel salon appointments online.
The system will provide a simple, user-friendly interface salon owner, stylists and clients.
The system will support stylist selection and service-based booking for better customer experience.
FYDP Objectives
The system will automate the salon appointment process and will reduce manual mistakes.
The system will design an easy and clear interface for stylists, salon admin and clients.
All clients, stylists and payment records will be stored securely in a database.
It will provide a simple, affordable and efficient solution for salon management.
FYDP Success Criteria
The website will run smoothly without errors.
clients will be able to book, cancel and reschedule appointments successfully.
All clients and appointment data will be store securely in the database.
Email confirmation will be sent automatically for every appointment creation or cancellation.
Assumptions:
Users will have internet access to use the website.
Both salon admin and clients will register with correct information.
The server and database will be running and secure.
Users will have basic knowledge of using websites.
Risks & Obstacles
Client may enter incorrect information during booking or payment.
System performance may slow down if bookings or records become very large.
Risk of compatibility issues on different browsers or mobile devices.

Email notification may fail if server configuration changes.
Organization Address: FCIT, University of the Punjab, Lahore, Pakistan
Target End Users: Salon admin, Clients / Customers
Suggested Project Supervisor: Sir Hasan Raza
Approved By: Sir Hasan Raza
Date: October 16, 2025

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Chapter No 1

Project Proposal

1.1. Introduction

Traditional manual appointment booking at Beauty Blush Salon is time-consuming and error-prone. Appointments are handled through calls or in-person visits, which often causes scheduling conflicts and poor record management. This project will aim to replace the manual process with a web-based salon appointment booking system. Clients will be able to book appointments online, while the admin will manage services, stylists and appointments through a single platform. The system will improve efficiency, reduce errors and simplify salon appointment management.

1.2. Background

Single salon booking systems still manage their appointments manually. They take bookings through calls or also write them in registers. This often leads to confusion, double bookings and delays. Clients also cannot check available time slots or stylist details online and they have to wait for hours. There is a need for a simple digital system to manage clients more easily. The Beauty Blush Salon will help salon manage clients and appointments efficiently. Clients will be able to book services, choose stylists and make partial payments online (i.e., 100Rs). The system will make bookings more accurate make the process easier for clients and help the salon owner clearly see and control daily appointments.

1.3. Problem Statement

Beauty Blush Salon manage bookings manually using phone calls or registers. This manual process creates confusion, booking mistakes and delays. Salon admin face difficulties in managing appointments, confirming clients and checking payment status. Clients also experience problems such as late booking confirmation and unclear service updates. The lack of real-time availability reduces overall efficiency. Therefore, a simple online booking system is needed to make the booking process faster, more accurate and more convenient for both salon owner and clients.

1.4. Stakeholders & Interests

Stakeholder	Description	Interest
Salon Admin	Person who will manage services, stylists and payments.	Manage client appointments easily, avoid booking problems and save time.
Clients	People who will use salon services and book appointment online.	Choose services, select stylists, make partial payments and book appointment without calling.

Table 2: Stakeholder & Interest

1.5. Objectives:

- The Beauty Blush Salon system will aim to simplify the appointments booking process.
- It will reduce manual work and booking mistakes in the salon.
- It will improve customer convenience through online booking and payments.
- The system will help the salon admin to organize services and appointments easily.
- The system will improve overall communication between clients, stylist and salon admin.

1.6. Scope:

- The system will be developed for one single salon only.
- Clients will be able to register, login and update their profiles.
- Clients will be able to book appointments and make partial online payments.
- The system will allow the salon admin to manage services, stylists and appointments.
- All booking and payment data will be stored in the secure database.

1.7. Assumptions

- User will have internet access to use the website.
- Both salon admin and clients will login with correct information.
- The server and database will be running and secure.
- User will have basic knowledge of using website.

1.8. Risks

- User may enter incorrect information during booking or payment.
- System performance may slow down if bookings or records become very large.
- Risk of compatibility issues on different browsers or mobile devices.
- Email notifications may fail if server configuration changes.

1.9. Success Criteria

User Registration and Authentication
Service Selection and Appointment Booking
Partial Payment Confirmation
Appointment Scheduling and Management
Service and Stylist Management
Client Dashboard
Real-time Notifications
Roll-based Access Control
Data Privacy and Security
Responsive User Interface
Search and Filter Options
Error Handling and Recovery
Deployment Readiness and Testing

Table 3: Success Criteria

1.10. Tools, Libraries & Technologies

Tools, Libraries, And Technologies	Tools	Version	Rationale
	Visual Studio Code	1.90+	IDE for efficient coding
	MongoDB Compass	Latest	GUI tool for MongoDB management
	MS Word	2021	Documentation creation
	Draw.io	Latest	Used to design UML diagrams
	Figma	Latest	Used to design UI/UX mockups
	GitHub	Latest	Version control and collaboration
	Libraries	Version	Rationale
	Axios	Latest	Used to send requests from frontend to backend APIs

	Bootstrap	5.3+	Enables responsive web design
	Node mailer	Latest	Used to send booking confirmation notifications
	React Date picker	Latest	Allows clients to select appointment date & time easily
	Mongoose	8+	Used to define schemas and manage MongoDB data
	JSON Web Token	Latest	Used for secure user and admin authentication
	Technology	Version	Rationale
	MongoDB	6+	NoSQL database to store data
	Express.js	4.18+	Backend framework used to create APIs
	React.js	18+	Used to build interactive user interfaces and dashboards
	Node.js	18+	Runs JavaScript code on the server side
	HTML5	Latest	Define the structure of web pages
	CSS3	Latest	Styling and layout design
	JavaScript	ES6	Core scripting language

Table 4: Tools, Technologies & Libraries

1.11. Work Division

Sr. No	Roll Number	Name	Role Assignment & Work Division
1.	089351	Sahrish Yaseen	Planning, Analysis, Design, Coding, Testing, Deployment
2.	089286	Mahnoor	Planning, Analysis, Design, Coding, Testing, Deployment

Table 5: Project Team Members Work Division

1.12. Conclusion

The Beauty Blush Salon project will aim to improve the salon's appointment management process by replacing the traditional manual system with a centralized web-based solution. By digitizing activities such as service browsing, stylist selection, appointment booking and payment confirmation, the system will reduce manual work and avoid booking problems. The system will provide an organized and user-friendly platform for both clients and admin. It will improve efficiency, save time and ensure better appointment management. Overall, this project will deliver a reliable and modern solution that meets the needs of the salon and its clients.

Chapter No 2

Literature Review

2.1 Literature Survey

At Beauty Blush Salon, appointment booking is handled manually through calls and in-person visits. This manual process causes scheduling conflicts and makes record management difficult. Many existing online salon booking systems are designed for multiple salons and are complex for small businesses. This project will aim to provide a simple web-based appointment booking system focused only on Beauty Blush Salon to improve appointment management and efficiency.

2.2 Related Work:

Several salon booking systems in Pakistan like TONI&GUY, Depilex, Allenora Salon, Seema Beauty Parlour and Nabila Salon have been introduced to manage salon operations online. These systems provide basic functionalities such as service listings, portfolios and simple booking forms. However, most lack real-time available slots, stylist selection, automated booking confirmation and partial payment options. Their limited features and manual confirmation reduce user convenience and overall system efficiency. The proposed Beauty Blush Salon addresses these gaps by providing a fully automated, user-friendly platform for real-time scheduling, stylist selection and online partial payment.

2.3 Gap Analysis:

Salons	Online Booking System	Real time slot	Stylist Selection	Online payment	Client / Admin Dashboard
TONI&GUY [1]	✓	✗	✗	✗	✗
Depilex [2]	✓	✗	✗	✗	✗
Allenora Salon [3]	✗	✗	✗	✗	✗
Seema Beauty Parlour [4]	✓	✓	✗	✗	✓
Canvas Salon & Studio [5]	✓	✗	✗	✗	✗
Nabila Salon [6]	✗	✗	✗	✗	✗
Beauty Blush Salon	✓	✓	✓	✓	✓

Table 6: Gap Analysis

2.4 Summary

The literature review highlights the limitations of existing salon appointment systems, which often rely on manual processes or provide complex solutions designed for multiple salons. These systems lack simplicity, proper scheduling and efficient record management. Previous solutions address only partial needs, leaving gaps in usability and efficiency. The proposed system aims to overcome these challenges by offering a simple, salon-specific appointment booking system designed to meet the needs of Beauty Blush Salon and its clients effectively.

Chapter No 3

Software Requirements

Specification

3.1 Requirements Analysis

The Software Requirements Specification for the Beauty Blush Salon will explain all the requirements of the system. This system will help the salon work automatically without manual effort. It will manage client accounts, services, stylists and appointment in an organized way. The SRS clearly describes how the system will work, what features it must include and how each type of user will use the system.

3.2 User classes and characteristics

User Class	User Characteristics
Admin	Salon staff who will manage the web application and daily work. They will understand salon services and stylist details. They will be able to add, update and delete services and stylists, manage bookings and control application settings.
Client	People who will use the salon web application to view services and book appointments. They will be able to create an account, login, choose services and stylist, book appointment, pay the appointment fee and manage their profile.

Table 7: User classes & Characteristics

3.3 Requirement Identifying Technique

- **Use case Technique:** To identify the requirements for the Beauty Blush Salon Booking Web Application, the use case technique will be used. This technique is suitable for systems where users perform many actions such as booking appointments, manage services, making payments and updating profiles. Use cases will describe each user action step by step and will help define clear functional requirements for both clients and admins.
- **Storyboarding:** The storyboard technique will also be used to support requirement identification. A storyboard is a visual sequence of screens that will show how a user moves through the system. It will help to understand the user flow, such as how a client login, select services, book an appointment. Storyboarding will allow the development team to visualize interface steps and improve usability before development.

3.4 Functional Requirements

This section will outline the core system features of Beauty Blush Salon along with their related functional requirements. Each feature will describe the expected behavior of the system from the user's perspective.

FR-1: Client Registration

Identifier	FR-1
Title	Client Registration
Requirement	The system will allow clients to create an account by providing name, email, password, confirm password and contact number.
Source	Client
Inputs	Name, Email, Password, confirm Password and Contact Number
Destination	Database
Outputs	The system will show your account has been created successfully.
Rationale	Registration is required so client can book appointments.
Business Rule (if required)	Email must be unique
Dependencies	None
Priority	High

Table 8:Function Requirements Specification for Client Registration

FR-2: Client Login

Identifier	FR-2
Title	Client Login
Requirement	The system will allow clients to login using registered email and password.
Source	Client
Inputs	Email, Password
Destination	Client Dashboard Screen
Outputs	Login Successful or invalid username or password.

Rationale	Login is required to access the client dashboard
Business Rule (if required)	Only registered users can login.
Dependencies	FR-1
Priority	High

Table 9: Function Requirements Specification for Client Login

FR-3: Forgot Password

Identifier	FR-3
Title	Forgot password
Requirement	The system will allow clients and admin to reset their password by verifying their registered email, receiving reset link and creating a new password.
Source	Client and Admin
Inputs	Email, New Password, Confirm New Password
Destination	Database
Outputs	Password Reset Successfully
Rationale	It will help users when they forgot their password.
Business Rule (if required)	Reset link must be valid and new password must match.
Dependencies	FR-1
Priority	High

Table 10: Function Requirements Specification for Forgot Password

FR-4: View Services

Identifier	FR-4
Title	View Services
Requirement	The system will allow clients to view all available services with price and duration.
Source	Client
Inputs	None

Destination	Services Page
Outputs	List of services with price, duration and description.
Rationale	Client must review services before booking.
Business Rule (if required)	Services data must be accurate and updated
Dependencies	FR-2
Priority	Medium

Table 11: Function Requirements Specification for View Services

FR-5: View Stylists

Identifier	FR-5
Title	View Stylists
Requirement	The system will allow clients to view stylist details including specialization and experience.
Source	Client
Inputs	None
Destination	Stylist list screen
Outputs	List of available stylists
Rationale	Client select stylist before booking
Business Rule (if required)	Stylists must be shown according to service availability
Dependencies	FR-2
Priority	Medium

Table 12: Function Requirements Specification for View Stylist

FR-6: Book Appointment

Identifier	FR-6
Title	Book Appointment
Requirement	The system will allow clients to book an appointment by selecting service, stylist, date and time and making appointment fee.

Source	Client
Inputs	Services, Stylist, Date and Time, Payment
Destination	Database
Outputs	Your appointment has been booked successfully
Rationale	This is the main function of the system.
Business Rule (if required)	Appointment fee must be at least 100
Dependencies	FR-2,FR-4, FR-5
Priority	High

Table 13: Function Requirements Specification for Book Appointment

FR-7: Client Dashboard

Identifier	FR-7
Title	Client Dashboard
Requirement	The system will allow clients to view appointments, manage profile, cancel appointment, book new appointment and logout from the dashboard.
Source	Client
Inputs	None
Destination	Database screen
Outputs	Profile information and appointment lists
Rationale	Dashboard gives full control and easy access to client's data in one place.
Business Rule (if required)	Only logged-in clients can access the dashboard.
Dependencies	FR-2
Priority	High

Table 14: Function Requirements Specification for Client Dashboard

FR-8: Admin Login

Identifier	FR-8
-------------------	------

Title	Admin Login
Requirement	The system will allow admin to log into system by entering a valid email and password.
Source	Admin
Inputs	Email, Password
Destination	Admin Dashboard
Outputs	Login successful or invalid email or password
Rationale	Admin must access secure system features
Business Rule (if required)	Email and password must match the admin's correct login details
Dependencies	None
Priority	High

Table 15: Function Requirements Specification for Admin Login

FR-9: Admin Dashboard

Identifier	FR-9
Title	Admin dashboard
Requirement	The system will display all stylists, all bookings, today's booking, all services.
Source	Admin
Inputs	None
Destination	Dashboard screen
Outputs	Static cards with all bookings, all services, all stylists and today's bookings.
Rationale	Admin needs a quick overview of salon activity
Business Rule (if required)	Only logged-in admin can view dashboard
Dependencies	FR-8
Priority	High

Table 16: Function Requirements Specification for Admin Dashboard

FR-10: Manage Services

Identifier	FR-10
Title	Manage Services
Requirement	The system will allow admin to add, update and delete services. Updated services will be shown to clients automatically.
Source	Admin
Inputs	Service name, category, duration, price
Destination	Database
Outputs	Confirmation message for service added, updated or deleted.
Rationale	Salon must update services as needed
Business Rule (if required)	Service price and duration cannot be empty
Dependencies	FR-9
Priority	High

Table 17: Function Requirements Specification for Manage Services

FR-11: Manage Stylists

Identifier	FR-11
Title	Manage Stylists
Requirement	The system will allow admin to add, update, delete stylists and set their availability and specialization
Source	Admin
Inputs	Name, Specialization, Timing, Experience, Status
Destination	Database
Outputs	Confirmation message
Rationale	Admin must control which stylist is available for booking
Business Rule (if required)	Each stylist must have at least one specialization.

Dependencies	FR-9
Priority	High

Table 18: Function Requirements Specification for Manage Stylists

FR-12: Manage Appointments

Identifier	FR-12
Title	Manage Appointments
Requirement	The system will allow admin to view and update appointment status in the database.
Source	Admin
Inputs	Client name, service name, stylist name, date,time
Destination	Appointment display
Outputs	Appointment List
Rationale	Admin needs to find appointments quickly.
Business Rule (if required)	Appointment must be booked.
Dependencies	FR-6, FR-8, FR-9
Priority	High

Table 19: Function Requirements Specification for Manage Appointment

FR-13: Profile Management

Identifier	FR-13
Title	Profile Management
Requirement	The system will allow the admin to update their profile information and change their password.
Source	Admin
Inputs	Name, Email, Phone Number, New Password Confirm New password
Destination	Admin profiles section
Outputs	Updated profile confirmation

Rationale	Admin must maintain secure updated information
Business Rule (if required)	Password must meet minimum crieteria
Dependencies	FR-8
Priority	Medium

Table 20: Function Requirements Specification for Update Profile

FR-14: Logout

Identifier	FR-14
Title	Logout
Requirement	The system will allow the admin and client to securely log out of the system.
Source	Admin and client
Inputs	Logout button click
Destination	Login page
Outputs	Logged out successfully
Rationale	To protect system security
Business Rule (if required)	Must end session completely
Dependencies	FR-2, FR-8
Priority	High

Table 21: Function Requirements Specification for Logout

FR-15: Email Notification Automation

Identifier	FR-15
Title	Email Notification Automation
Requirement	The system will automatically send email notifications for booking confirmation and cancellation
Source	System
Inputs	User email, message content

Destination	Email Service
Outputs	“Email Sent Successfully”
Rationale	Keeps clients informed about their bookings.
Business Rule (if required)	Email must be sent after booking data is saved
Dependencies	FR-6
Priority	Medium

Table 22: Function Requirements Specification for Email Notification Automation

FR-16: Appointment Reminder Notification

Identifier	FR-16
Title	Appointment Reminder Notification
Requirement	The system will automatically send reminder notifications to client before their appointment time.
Source	System
Inputs	Client email, appointment date and time
Destination	Email notification service
Outputs	Reminder sent
Rationale	Helps client reminder their appointments and reduce no show.
Business Rule (if required)	Reminder must be sent for confirmed appointments only.
Dependencies	FR-6, FR-12
Priority	Medium

Table 23: Function Requirements Specification for Appointment Reminder Notification

3.5 Non-Functional Requirements

Non-functional requirements describe how the system should behave and what quality standards it must follow. These requirements do not describe features, but they describe the performance, reliability, security and quality of the software. They must be specific, measurable and testable.

1. Reliability

The Beauty Blush Salon will maintain a Mean Time Between Failures (MTBF) of at least 500 hours under normal usage. The system will recover from any crash within 30 seconds after restart (MTTR). Incremental backups of user data will be performed automatically every Sunday. The system will ensure at least 98% data integrity after recovery and all system errors will be logged for monitoring and correction.

2. Usability

The Beauty Blush Salon System will be designed to be simple and easy for all users. Clients will be able to book an appointment in no more than 4 taps. Clear buttons, readable text and on-screen guidance messages will help users to understand each step and avoid mistakes. The website will work smoothly on mobile devices with large buttons and a clean layout and will achieve at least 85% satisfaction during testing.

3. Performance

The Beauty Blush Salon Booking System must respond within fixed measurable limits. All main pages (Home, services, stylists, booking, dashboard) must load within 3 seconds. Appointment booking steps must complete within 6 seconds. Payment confirmation message must appear within 4 seconds after the user submits the payment. All database operations such as booking or updating profiles must be completed within 2 seconds.

4. Security

The Beauty Blush Salon Booking System must protect all user and admin information from unauthorized access. The system should block all invalid login attempts and lock an account after 5 failed tries. Booking, profile and payment data must remain protected.

3.6 External Interface Requirements

The Beauty Blush Salon Booking System will work smoothly on mobiles, tablets and computers. It will connect securely with an online payment service for advance payments and

show confirmation message. The system will send booking and cancellation emails to clients. It will store and retrieve client, stylist, service and appointment data from the database. The web app will run correctly on common browsers like chrome and edge and all data will be protected.

1. User Interfaces Requirements

The Beauty Blush Salon Booking System will have a clean and simple interface that will be easy for all users to understand. All screens will follow a consistent design so users do not get confused. The Beauty Blush Salon Web App will work properly on both mobile phones and computers. The system will follow these user interface rules:

- Each page will display a clear navigation bar with Home, Services, Stylist, Gallery and login options.
- All buttons will use simple and clear labels such as Book Appointment, Add, Save, Update and Cancel.
- The form fields will be shown in a clear order so users can fill the form easily without confusion.

2. Software Interfaces

The Beauty Blush Salon Booking System will use Node.js and Express.js as the backend technology and MongoDB as the database. The system will run on a server environment supported by Node.js. HTML, CSS, JavaScript will be used for the front-end design. Nodemailer will be used to send email notifications such as booking confirmation and cancellation. The system will support common web browser and ensure smooth communication between the application and the database.

3. Hardware Interfaces

The Beauty Blush Salon Booking System will work on common devices such as laptop, desktop, tablets and smartphones. Users will only need a basic device with a web browser and an internet connection to access the system and perform bookings or payments. The server side of the system will run on a machine that supports Node.js and MongoDB. No special hardware will be required.

4. Communications Interfaces

The Beauty Blush Salon Booking System will use the server to handle communication between the application and the MongoDB database. Nodemailer will be used for sending booking confirmation and password reset emails. The system will work on major browsers

such as chrome, edge and firefox. An active internet connection is required for system usage and automated emails will be sent shortly after user actions.

3.7 Use case Analysis

1. Use Case #1 (Register Account – UC1)

UC Identifier	UC1
UC Name	Register Account
Requirements Traceability	FR-1
Purpose	The system will allow a new client to create an account to use the system.
Priority	High
Preconditions	Client must have a valid email. Client must not be already registered.
Post conditions	New client account will be created and stored in the system.
Actors	Client
Extends	None
Main Success Scenario	Client will open registration page. System will ask for name, email, password, confirm password and phone number. Client will enter all required details. System will check if email is unique and will ensure it's not already in the system. System will create new account. System will show registration successful message.
Alternate Flows	If the email is already registered, the system will show an error message. If password does not match, the system will show a password mismatch.
Exceptions	If the system or database error occurs, the account will not be created.
Includes	Name, Email, Password, Confirm Password and Phone Number

Table 24: Use case Description for the Register Account

2. Use Case #2 (Login – UC2)

UC Identifier	UC2
UC Name	Login

Requirements Traceability	FR-2
Purpose	The system will allow client to access their account.
Priority	High
Preconditions	Client must be already registered.
Post conditions	Client will be logged in and dashboard will appear.
Actors	Client
Extends	None
Main Success Scenario	Client will open login page. System will ask for email and password. Client will enter both email and password. System will verify credentials. System will log client in and will redirect to dashboard.
Alternate Flows	If email is not found, the system will show “No account exists”. If the password is incorrect, the system will show an error message.
Exceptions	If a server error occurs, the system will not verify email or password.
Includes	Email and Password

Table 25: Use case Description for Login

3. Use Case #3 (Forgot Password – UC3)

UC Identifier	UC3
UC Name	Forgot Password
Requirements Traceability	FR-3
Purpose	The system will allow client to reset their password.
Priority	High
Preconditions	Client must have valid registered email.
Post conditions	Password will be updated
Actors	Client
Extends	None
Main Success Scenario	Client will click forgot password. System will ask for email and client will enter registered email. System will generate and send reset link to the client’s email.

	Client will open email and click the reset link. System will verify the link and will show the reset password page. Client will enter and will confirm new password. System will update the password and will show success message.
Alternate Flows	If reset link is invalid or expired, the system will show invalid or expired reset link message.
Exceptions	If a network issue occurs, the reset link will not be sent. If a server error occurs, the password will not be updated.
Includes	Enter Email, Reset link, New password, Confirm New Password.

Table 26: Use case Description for Forgot Password

4. Use Case #4 (View Services – UC4)

UC Identifier	UC4
UC Name	View Services
Requirements Traceability	FR-4
Purpose	The system will allow the client to see all salon services with their prices and duration.
Priority	Medium
Preconditions	Client must be on the web app. Services must exist in the database.
Post conditions	Service list will be displayed successfully.
Actors	Client
Extends	None
Main Success Scenario	Client will click view services. System will fetch list of services from database. System will display services list. Client will be able to select any one service. System will show service categories for the selected type. System will show each service with its price and duration. Client will choose any service to continue to stylist selection.
Alternate Flows	If no services are available, the system will show no services found message.
Exceptions	If a database error occurs, services will not be loaded.

Includes	None
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Table 27: Use case Description for View Services

5. Use Case #5 (View Stylists – UC5)

UC Identifier	UC5
UC Name	View Stylists
Requirements Traceability	FR-5
Purpose	The system will allow client to see all available stylists and their specialization.
Priority	Medium
Preconditions	Stylists must be added by admin. Client will open the stylist page using view stylist or through service selection.
Post conditions	Stylist list will be displayed. Client will select a stylist.
Actors	Client
Extends	None
Main Success Scenario	Client will click on stylists. System will retrieve stylist data. System will display image, name, specialization and experience. Client will select a stylist to continue.
Alternate Flows	If no stylists are available, the system will show no stylist available message.
Exceptions	If database connection fails, stylist data will not be retrieved.
Includes	None

Table 28: Use case Description for the View Stylists

6. Use Case #6 (Book Appointment – UC6)

UC Identifier	UC6
UC Name	Book Appointment
Requirements Traceability	FR-6

Purpose	The system will allow client to book appointment by selecting service, stylist, date and time and make the payment.
Priority	Very High
Preconditions	Client must be logged in. services must exist. Stylist must be active. Time slots must be generated.
Post conditions	Appointment will be saved in database. Payment page will open automatically.
Actors	Client
Extends	None
Main Success Scenario	Client will choose service. Client will select stylist. System will show available time slots. Client will select date and time and will confirm. System will save appointment temporarily and will open payment page.
Alternate Flows	If the stylist is not available on the selected date, system will show alternative dates. If the selected time slot is already booked, the system will ask the client to choose another slot.
Exceptions	If a system error occurs, available time slots will not be fetched. If a network error occurs, the appointment will not be saved.
Includes	Select service, select stylist, select date and time, make payment.

Table 29: Use case Description for Book Appointment

7. Use Case #7 (Update Profile – UC7)

UC Identifier	UC7
UC Name	Update Profile
Requirements Traceability	FR-7
Purpose	The system will allow client to update personal information and change password.
Priority	Medium
Preconditions	Client must be logged in
Post conditions	New profile information will be stored.
Actors	Client

Extends	Name, email, password and phone number
Main Success Scenario	Client will open profile setting. System will show current information. Client will update information. System will validate and will save changes.
Alternate Flows	If client enter invalid email format, system will show error message. If phone number is incorrect or invalid, system will show error message.
Exceptions	If database update fails, changes will not be saved. If network error occurs, system will not save changes temporarily.
Includes	None

Table 30: Use case Description for Update Profile

8. Use Case #8 (Logout – UC8)

UC Identifier	UC8
UC Name	Logout
Requirements Traceability	FR-14
Purpose	The system will end the user session safely.
Priority	High
Preconditions	Client must be logged in.
Post conditions	Client session will be destroyed
Actors	Client
Extends	None
Main Success Scenario	Client will click logout. System will end session. System will redirect to homepage
Alternate Flows	If session is already expired, system will redirect to login page with a timeout message.
Exceptions	If session not found, system will show “Session not found” error.
Includes	None

Table 31: Use case Description for Logout

9. Use Case #9 (Admin Login – UC9)

UC Identifier	UC9
UC Name	Admin Login
Requirements Traceability	FR-8
Purpose	The system will allow the admin to securely access the system using valid login email and password.
Priority	High
Preconditions	Admin must already have an account. Admin must remember correct email and password.
Post conditions	Admin will be logged in successfully. System will redirect admin to dashboard.
Actors	Admin
Extends	None
Main Success Scenario	Admin will open login page. System will display email and password fields. Admin will enter valid email and password. System will verify email and password. System will log in admin. System will redirect to dashboard.
Alternate Flows	If admin enters invalid email, system will show “Email not found”. If admin enters wrong password, system will show “Incorrect password”.
Exceptions	If database connection fails, system will not be able to verify credentials. If server error occurs, login attempt will fail.
Includes	Email and password

Table 32: Use case Description for Admin Login

10. Use Case #10 (Forgot Password – UC10)

UC Identifier	UC10
UC Name	Forgot Password
Requirements Traceability	FR-3

Purpose	The system will help admin recover login access by resetting forgotten password.
Priority	Medium
Preconditions	Admin must have a registered email in system.
Post conditions	Password will be updated successfully. Admin will be able to login with new password.
Actors	Admin
Extends	None
Main Success Scenario	Admin will click forgot password. System will ask for email and client enters registered email. System will generate and send reset link to the admin's email. Admin opens email and click the reset link. System will verify the link and will show the reset password page. Admin will enter and will confirm new password. System will update the password and will show success message.
Alternate Flows	If reset link is invalid or expired, the system will show invalid or expired reset link message. If admin email is not registered, system will show "Email not registered".
Exceptions	If network issue occurs, reset link will not be sent. If server error occurs, password will not be updated temporarily.
Includes	Enter Email, Reset link, New password, Confirm New Password.

Table 33: Use case Description for Forgot Password

11. Use Case #11 (View Dashboard – UC11)

UC Identifier	UC11
UC Name	View Dashboard
Requirements Traceability	FR-9
Purpose	The system will allow admin to view overall summary of system activities.
Priority	High
Preconditions	Admin must be logged in.
Post conditions	Dashboard details will be displayed successfully.
Actors	Admin

Extends	None
Main Success Scenario	Admin will log in. System will redirect to dashboard. Dashboard will show counts (All appointments, All services, All stylist and today's booking)
Alternate Flows	If dashboard widgets fail to load, system will show fallback message.
Exceptions	If server is unavailable, dashboard will not be displayed. If database connection fails, dashboard counts will not load.
Includes	None

Table 34: Use case Description for Admin Dashboard

12. Use Case #12 (Manage Services – UC12)

UC Identifier	UC12
UC Name	Manage Services
Requirements Traceability	FR-10
Purpose	The system will allow admin to manage all salon services such as add, delete and update.
Priority	High
Preconditions	Admin must be logged in. Admin must have access to dashboard.
Post conditions	Service list will be updated in system. Changes will become visible to clients immediately.
Actors	Admin
Extends	View services Add Service, Update Service and Delete Service
Main Success Scenario	Admin will open Manage Services page. System will display all available services. Admin will select an action (add/update/delete). System will perform selected action. System will update and will show new service list.
Alternate Flows	If admin tries to add duplicate service, system will show “Service already exists”.

	If admin deletes a service with ongoing appointments, system will show warning.
Exceptions	If database error occurs while saving, changes will not be applied. If network fails during update request, action will be aborted.
Includes	None

Table 35: Use case Description for Manage Services

13. Use Case #13 (Manage Stylists – UC13)

UC Identifier	UC13
UC Name	Manage Stylists
Requirements Traceability	FR-11
Purpose	The system will allow admin to add, update and delete stylists.
Priority	High
Preconditions	Admin must be logged in. Stylist table must exist in database.
Post conditions	Stylist information will be added, deleted an updating. Status will be updated properly.
Actors	Admin
Extends	View Stylist, Add Stylist, Update Stylist, Delete Stylist
Main Success Scenario	Admin will open manage stylists page. System will display list of all stylists. Admin will select Add, Update or Delete System will perform selected action. System will show confirmation,
Alternate Flows	If no stylist exists, system will show “No data available” message. If stylist has pending appointments, system will show warning before deletion or status change.
Exceptions	If database error occurs while saving, action will fail. If invalid fields are entered, system will show validation errors.
Includes	None

Table 36: Use case Description for Manage Stylists

14. Use Case #14 (Manage Appointments – UC14)

UC Identifier	UC14
UC Name	Manage Appointments
Requirements Traceability	FR-12
Purpose	The system will allow admin to view and update status of appointment.
Priority	High
Preconditions	Admin must be logged in. Appointment data must exist.
Post conditions	Appointment status will be updated.
Actors	Admin
Extends	View Appointment and update status.
Main Success Scenario	Admin will open Manage Appointment page. Admin will apply filters. System will display filtered appointments. Admin will view appointments. Admin will update status.
Alternate Flows	If no appointment found for applied filters, system will show “No results found” message.
Exceptions	If database connection fails, appointments will not load. If appointment has already been updated, system will show “Appointment already updated” message.
Includes	None

Table 37: Use case Description for Manage Appointment

15. Use Case #15 (Update Profile – UC15)

UC Identifier	UC15
UC Name	Update Profile
Requirements Traceability	FR-13
Purpose	The system will allow admin to update profile and change password.
Priority	Medium
Preconditions	Admin must be logged in.

Post conditions	Profile will be updated successfully.
Actors	Admin
Extends	Name, Email and Change password
Main Success Scenario	Admin will open update profile page. Admin will change her name, email and password. System will validate data.
Alternate Flows	If password and confirm password not match, system will show “password mismatch” message If email will already in use, system will show “Email already in use” message.
Exceptions	If password is weak, system will reject update. If database error occurs, profile changes will not be saved.
Includes	None

Table 38: Use case Description for Update Profile

16. Use Case #16 (Logout – UC16)

UC Identifier	UC16
UC Name	Logout
Requirements Traceability	FR-14
Purpose	The system will securely end the admin session.
Priority	High
Preconditions	Admin session must be active.
Post conditions	Session will be destroyed and redirected to login page.
Actors	Admin
Extends	None
Main Success Scenario	Admin will click logout. System will clear session tokens. System will redirect to login page.
Alternate Flows	If session is already expired, system will redirect to login page with timeout message.
Exceptions	If session not found system will show “Session not found” error.
Includes	None

Table 39: Use case Description for Logout

17. Use Case Diagram

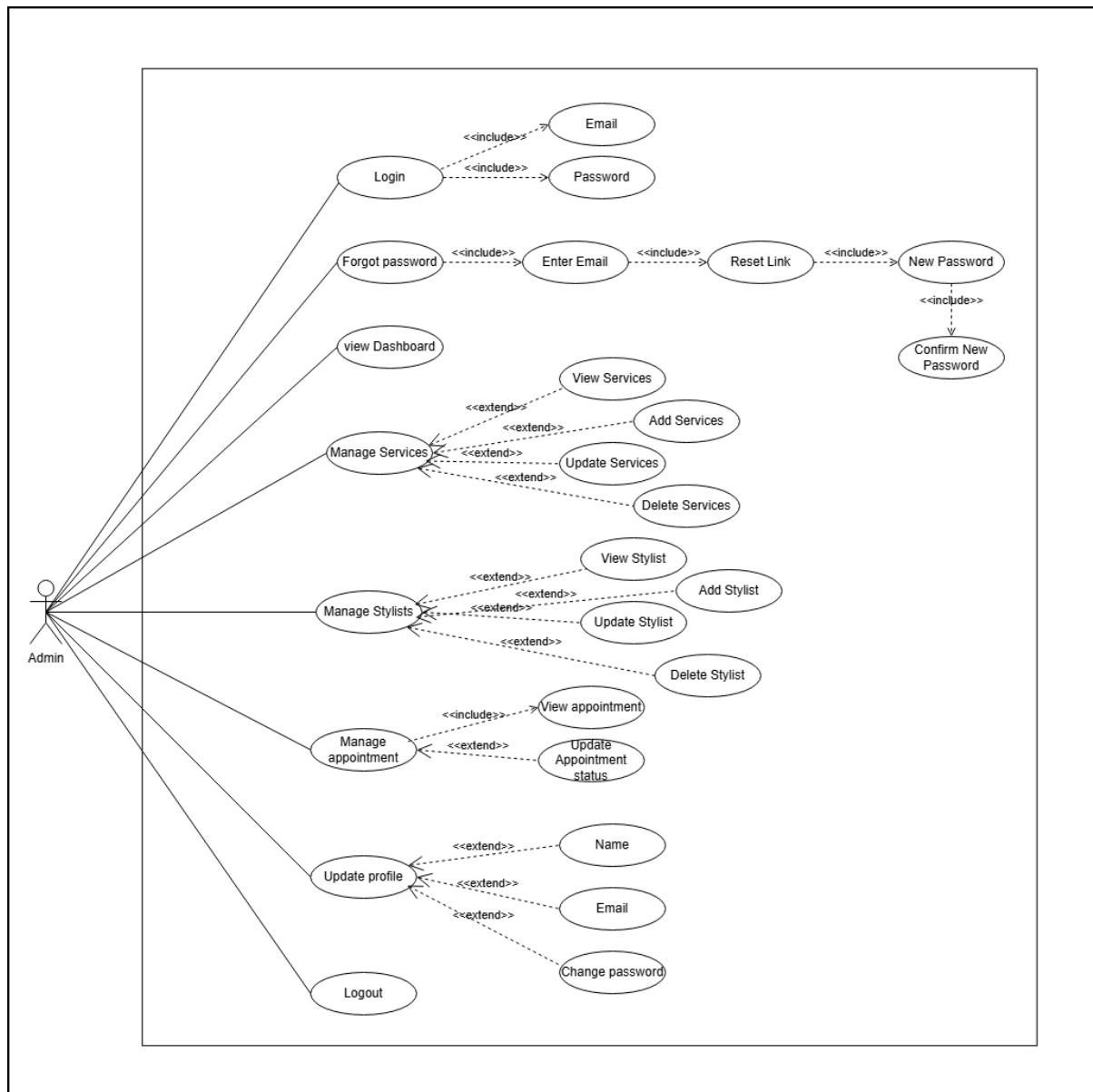


Figure 1: Admin use case diagram

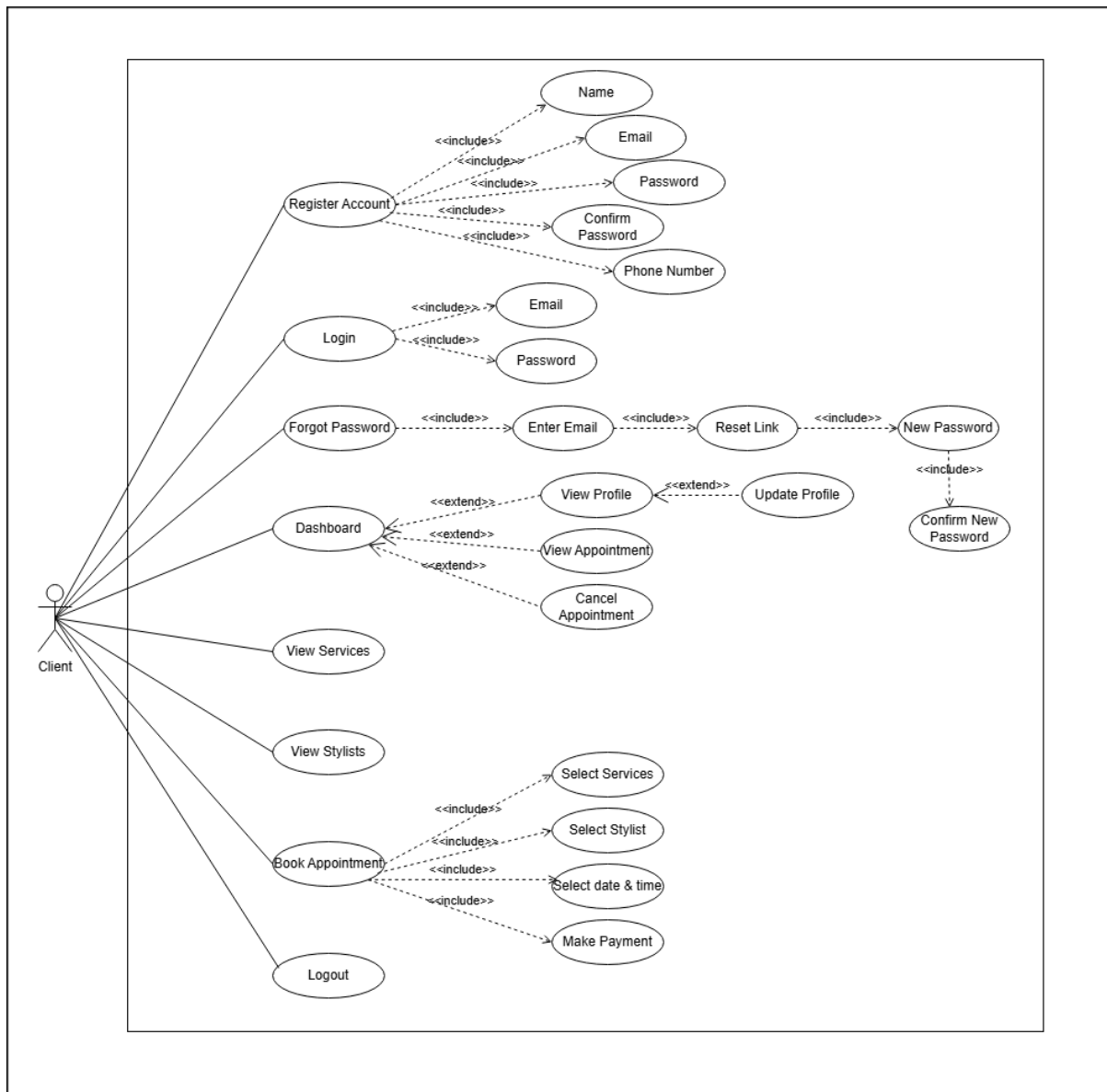


Figure 2: Client use case diagram

3.8 Storyboards

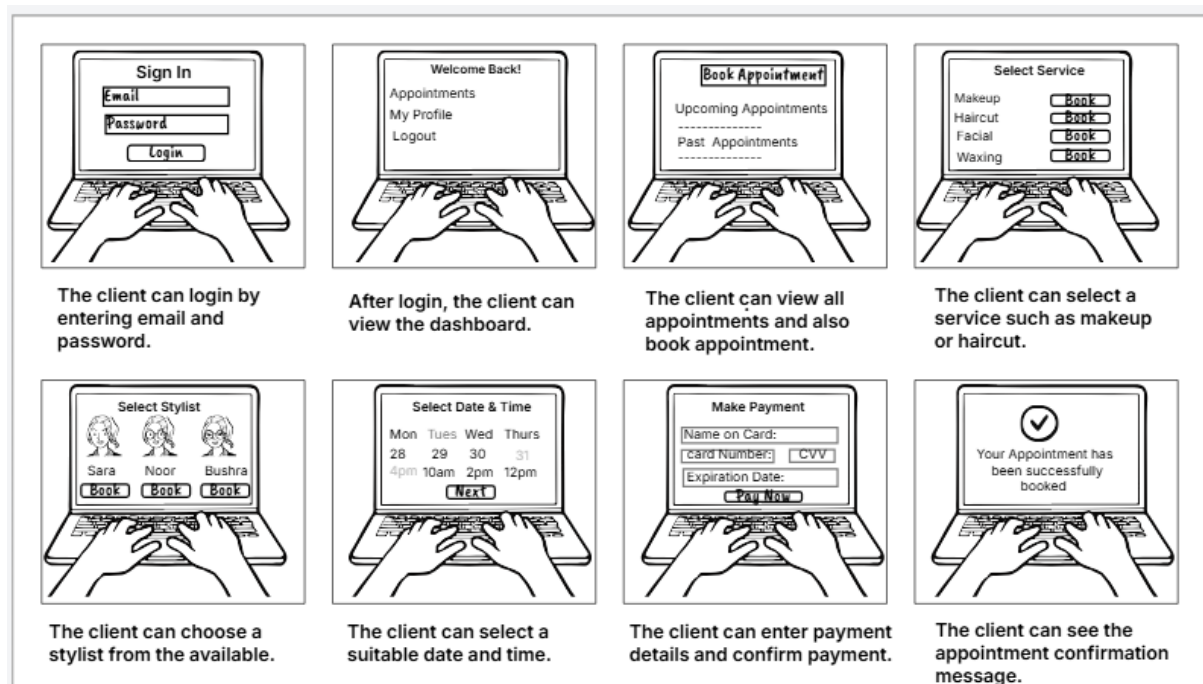


Figure 3: Storyboard for Book Appointment

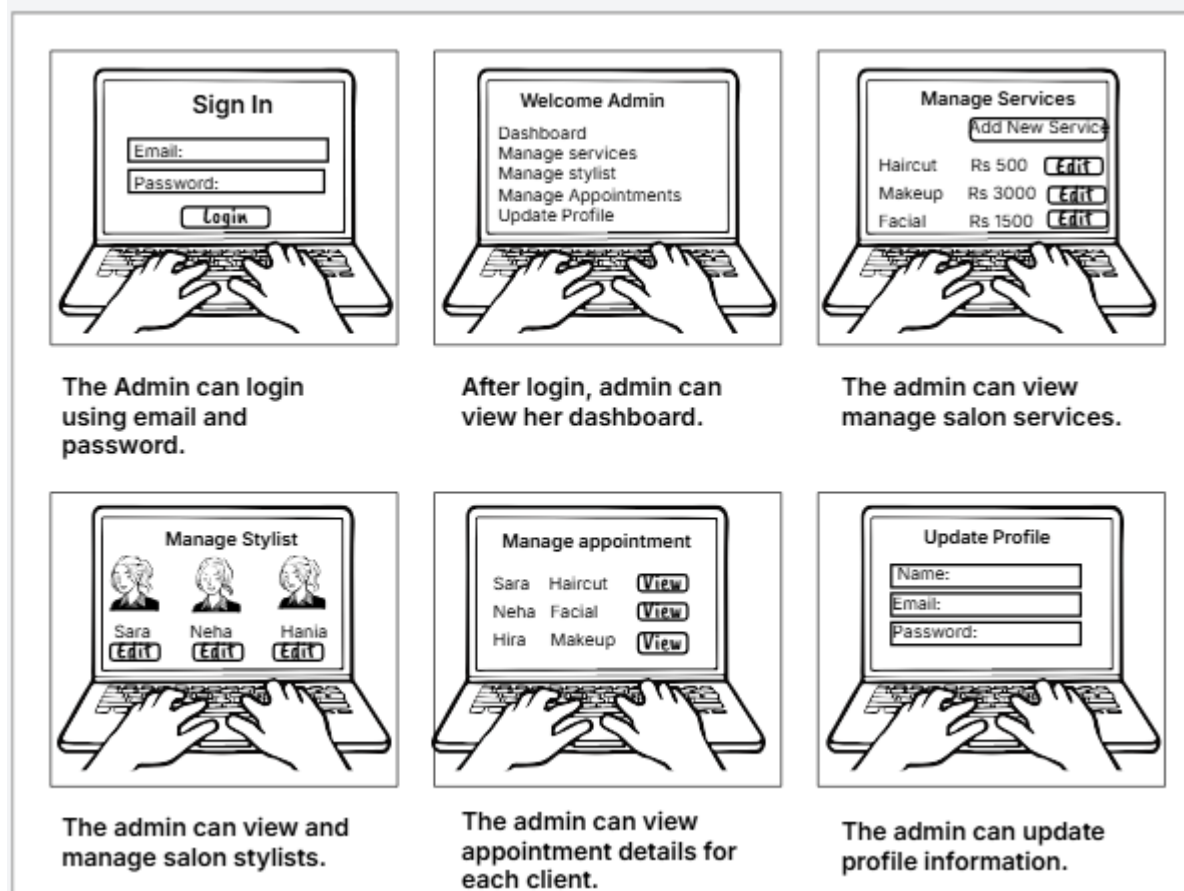


Figure 4: Storyboard for Admin Manage system

3.9 Summary

The requirements identification and use case analysis for the Beauty Blush Salon Booking System will focus on understanding client and admin needs. Functional and non-functional requirements will be gathered and documented to define how the system should work. User-system interactions will be identified through use cases. These requirements will be validated to match project objectives. This process will help to ensure the system works efficiently and simplifies the salon appointment booking and management process.

Chapter No 4

Software Design Specification

4.1 System Design

The Beauty Blush Salon appointment booking system will work as a standalone web-based application. It will be accessed through a web browser and will require an internet connection for Admin and Clients. The system will allow clients to register, login, book appointment and make payments online. Admin will manage services, stylists and appointments.

Design Constraints:

- The system must support multiple users at the same time without performance issues.
- It must be accessible on both desktop and mobile browsers using responsive design.
- Hosting will be done on a limited web server.
- Only web-based access is provided; no mobile application is included.
- Secure login and data protection will be ensured.
- System updates will be managed using version control tools.

4.2 Design Considerations

Assumptions and Dependencies:

- Users will have internet access and basic knowledge of using web applications.
- The system will depend on a web server and a database.
- A notification service will be used for sending confirmations and updates.
- Payments will be processed through online methods support by the system.

Limitations:

- Offline usage is not supported.
- The system is only available as a web application.
- User roles are predefined and cannot be customized.
- Advanced features such as biometric authentication or GPS tracking are not included.

Risks:

- **Data Security Risks:** Reduced by implementing secure login and role-based access control.
- **Performance Issues:** Minimized by using efficient development practices.
- **User Adoption Challenges:** Addressed by providing a simple and user-friendly interface.
- **System Downtime:** Reduced by proper hosting and regular system maintenance.

4.3 Requirements Traceability Matrix

Requirement ID	Scope	Requirement Description	Design Specification
R1	Client Registration	The system will allow the client to create an account by providing name, email, password, confirm password and phone number	Component “User Registration and Account Management Module”
R2	Client Login	The system will allow registered clients to log in using their email and password.	Component “Authentication Module”
R3	Forgot Password	The system will allow clients and admin to reset their password via registered email verification	Component “Password Reset Module”
R4	View Services	The system will allow clients to view available salon services, including prices and duration.	Component “Services Display Module”
R5	View Stylists	The system will allow clients to view available stylists and their specialization.	Component “Stylist Display Module”
R6	Book Appointment	The system will allow clients to book appointments by selecting service, stylist, date and time and make payment.	Component “Appointment Booking Module”
R7	Client Dashboard	The system will allow clients to manage their profile, view appointments and logout from the dashboard.	Component “Client Dashboard Module”
R8	Admin Login	The system will allow the admin to login securely with a valid email and password.	Component “Admin Authentication Module”
R9	Admin Dashboard	The system will display an overview of salon activity.	Component “Admin Dashboard Module”

R10	Manage services	The system will allow the admin to add, update and delete services.	Component “Service Management Module”
R11	Manage Stylists	The system will allow the admin to add, delete and manage the availability of stylists.	Component “Stylist Management Module”
R12	Manage Appointments	The system will allow the admin to view and manage appointment status.	Component “Appointment Management Module”
R13	Admin Profile Management	The system will allow admin to update their profile and change their password.	Component “Admin Profile Management Module”
R14	Session Management	The system will allow both admin and client to securely log out, destroyed their session.	Component “Session Management Module”
R15	Email Notification	The system will automatically send email notifications for booking confirmations or cancellation.	Component “Email Notification Module”
R16	Appointment Reminder	The system will automatically send reminder notifications to clients before their appointment.	Component “Reminder Notification Module”

Table 40: Requirement Traceability Matrix

4.4 Design Models

Design models will describe the internal structure of the Beauty Blush Salon system. They will show classes, interactions and state changes of the system.

1. Design Class Diagram (DCD)

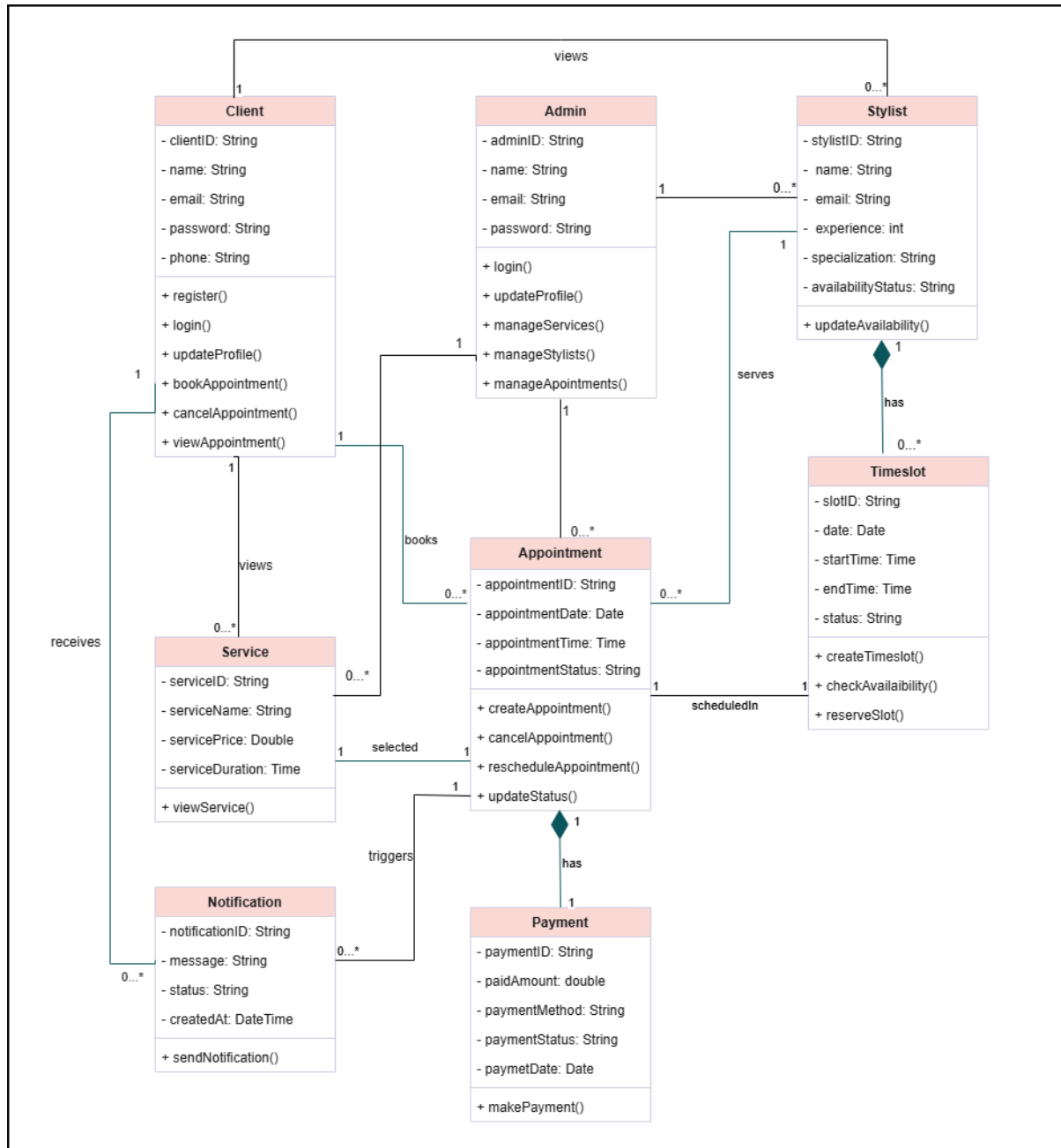


Figure 4: Class diagram

2. Interaction Diagram (Either sequence or collaboration)

The Interaction Diagram will show how system components will interact during different use cases. Sequence diagrams will be created for appointment booking, payment processing and appointment cancellation.

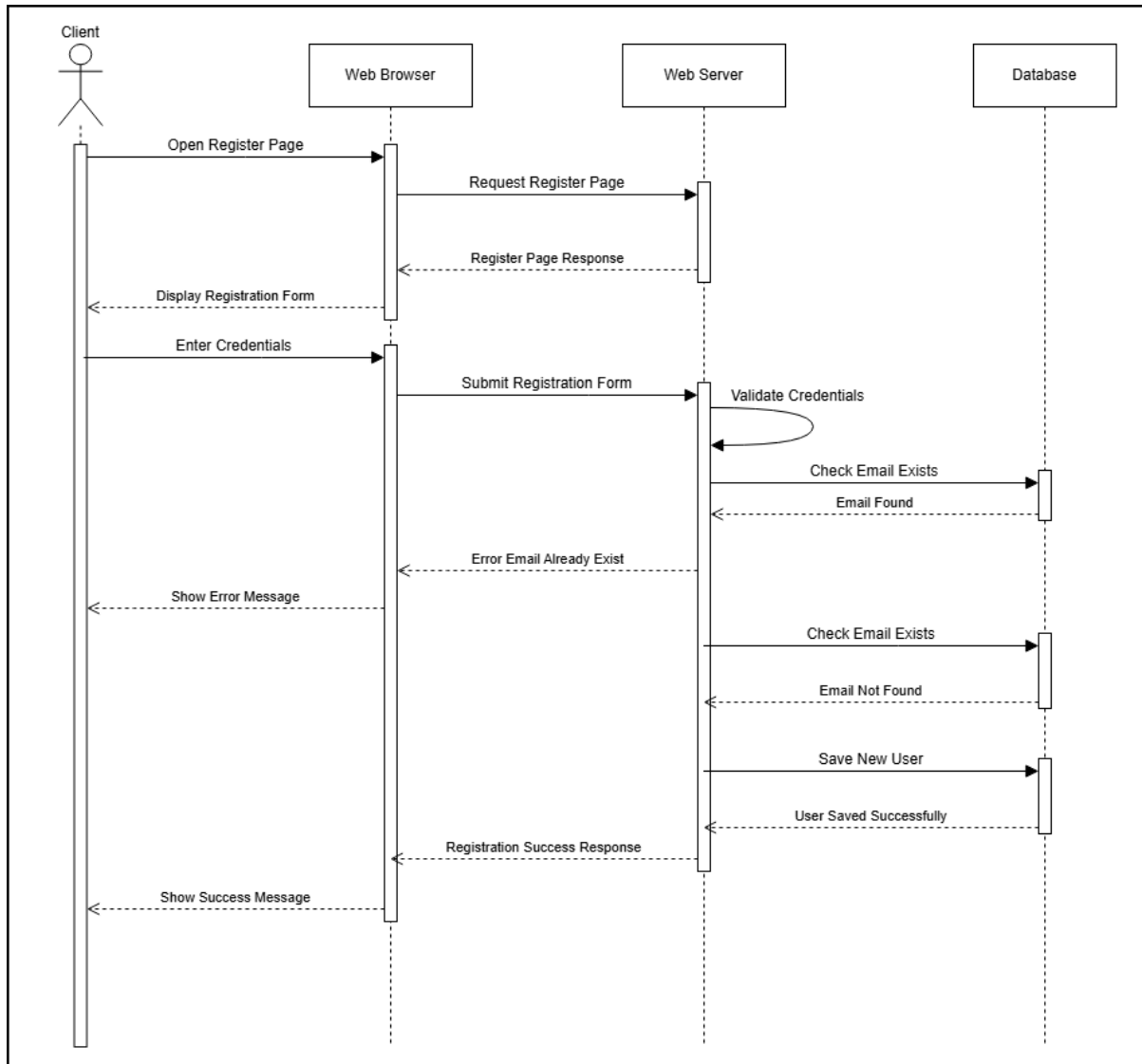


Figure 5: Client Register Sequence Diagram

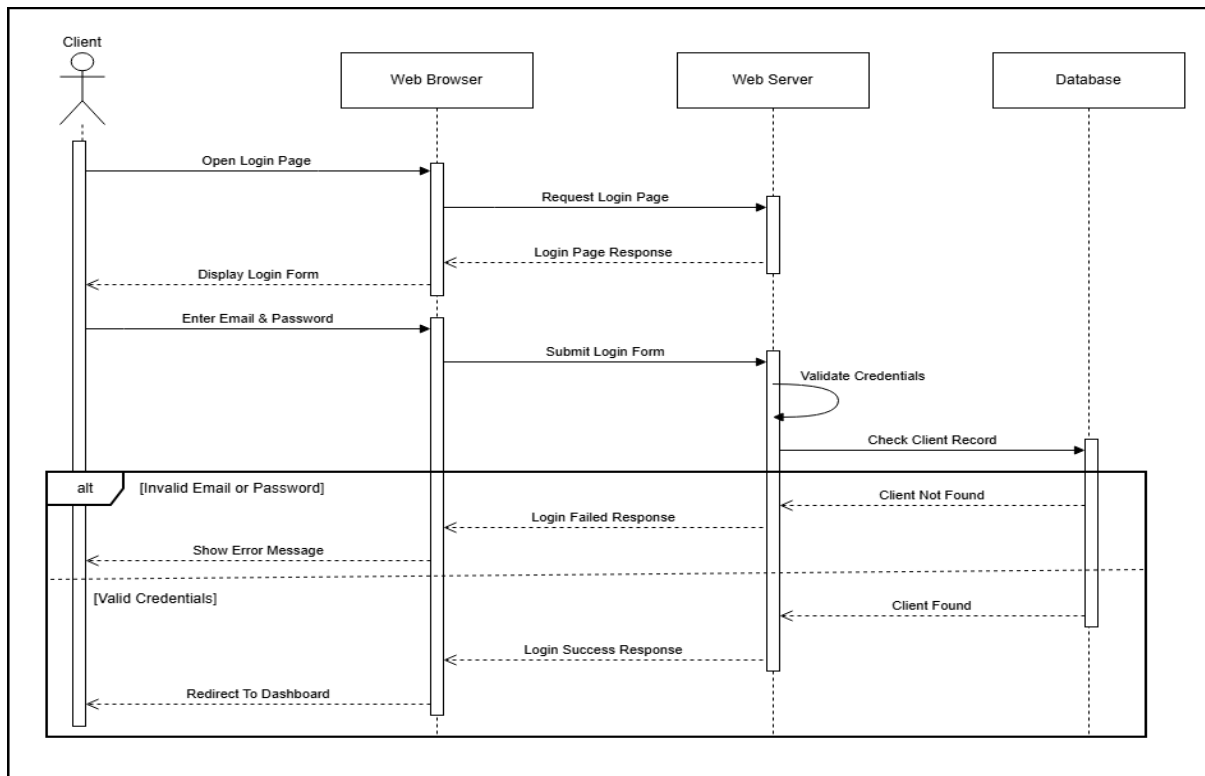


Figure 6: Client Login Sequence Diagram

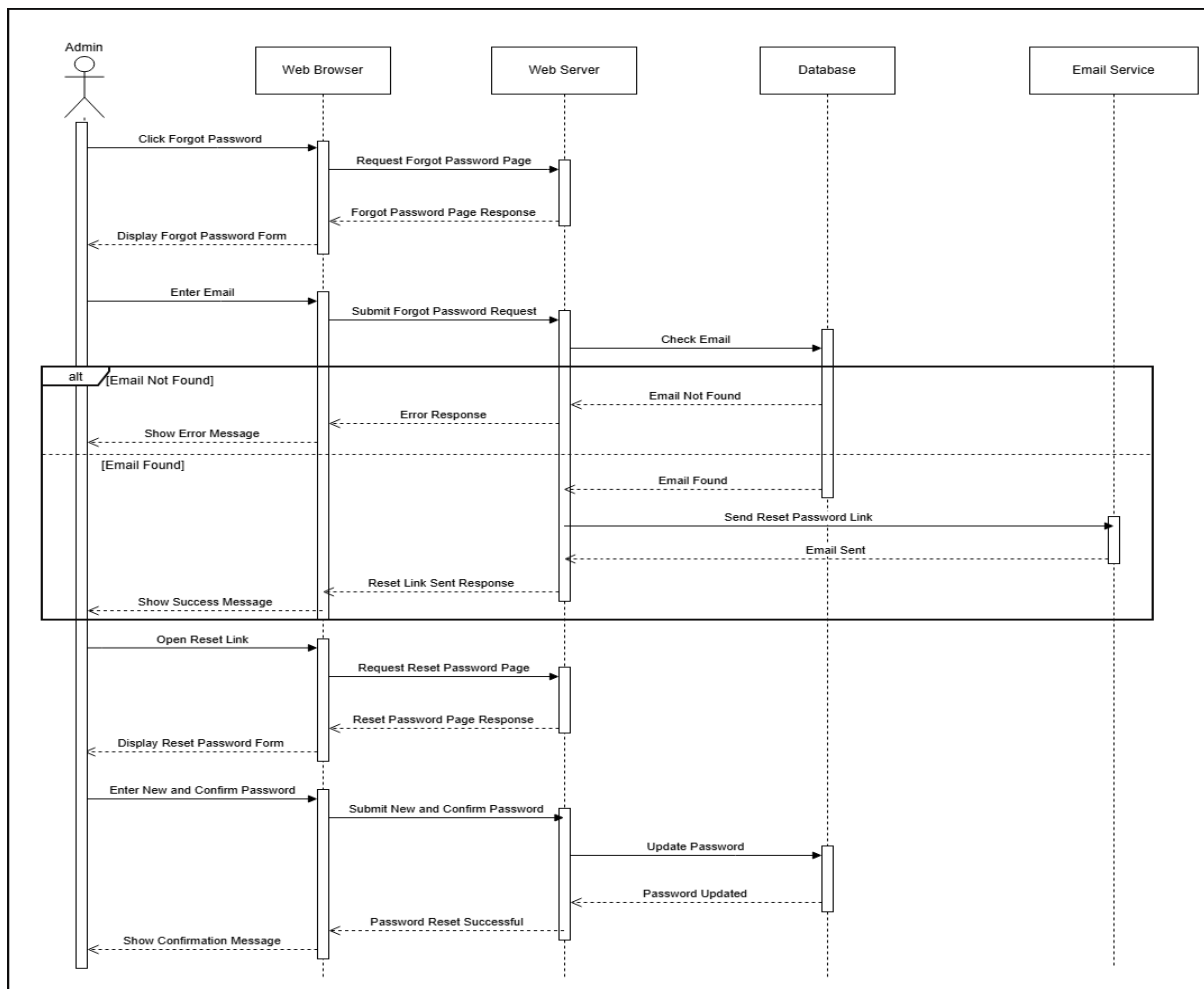


Figure 7: Client Forgot Password Sequence Diagram

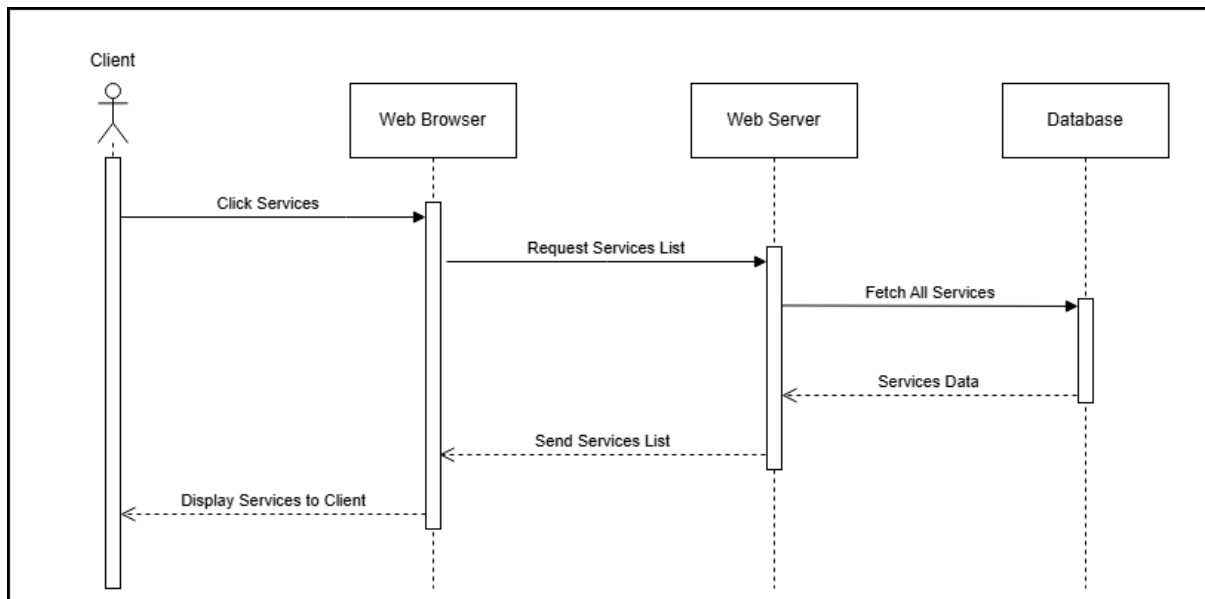


Figure 8: Client View Services Sequence Diagram

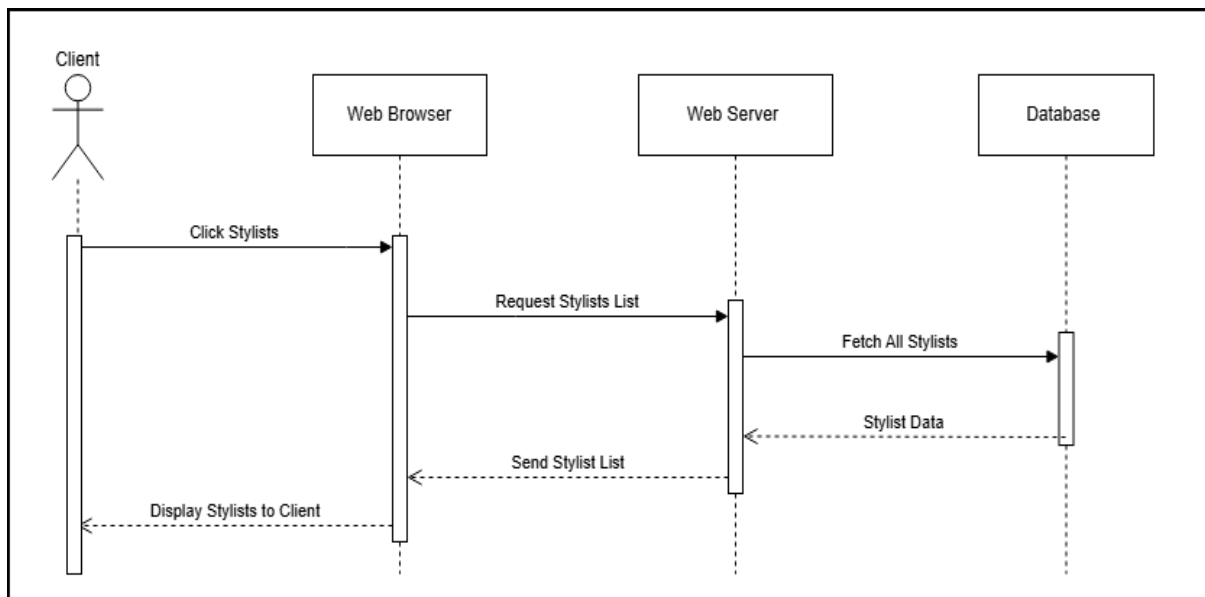


Figure 9: Client View Stylist Sequence Diagram

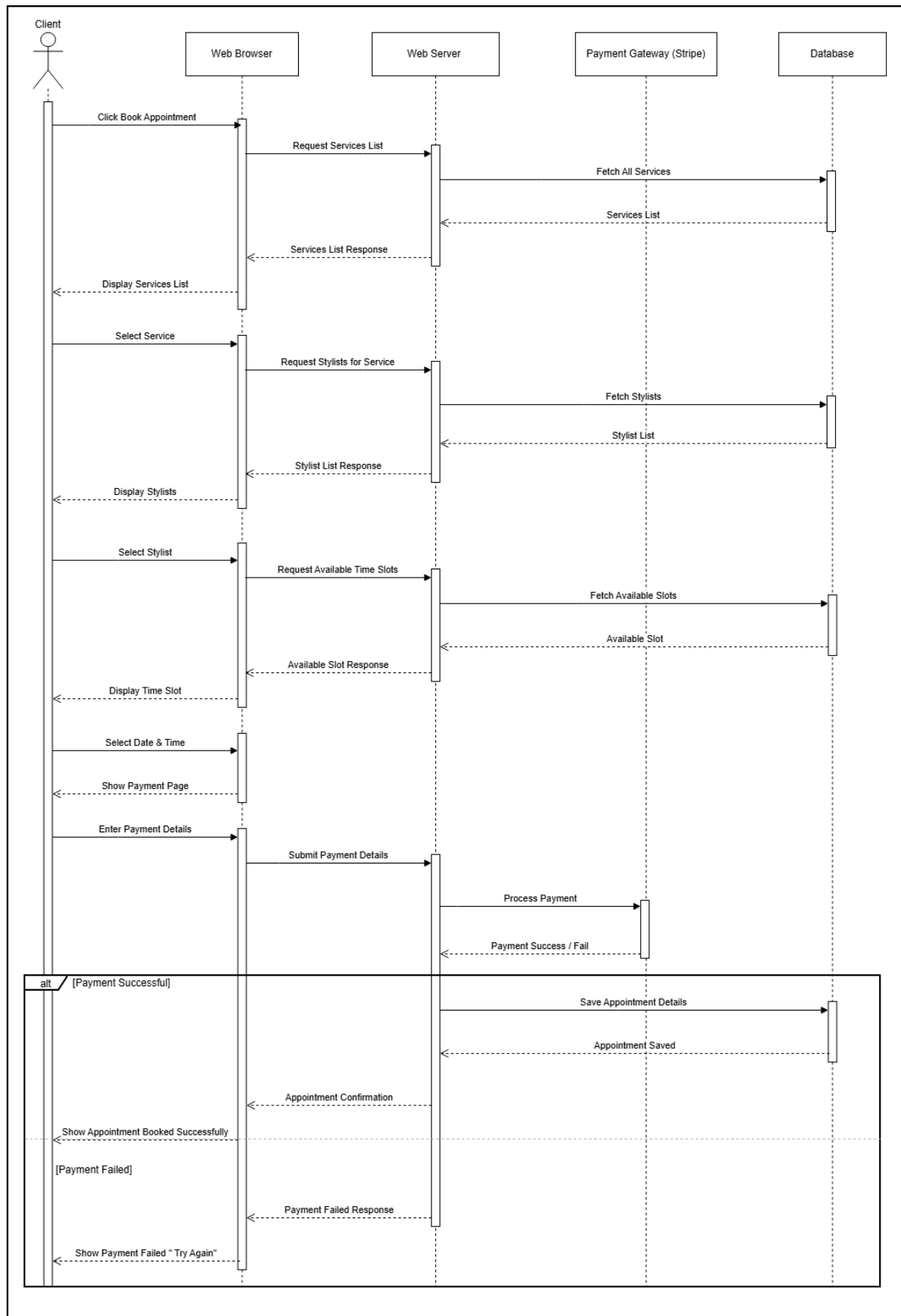


Figure 10: Book Appointment Sequence Diagram

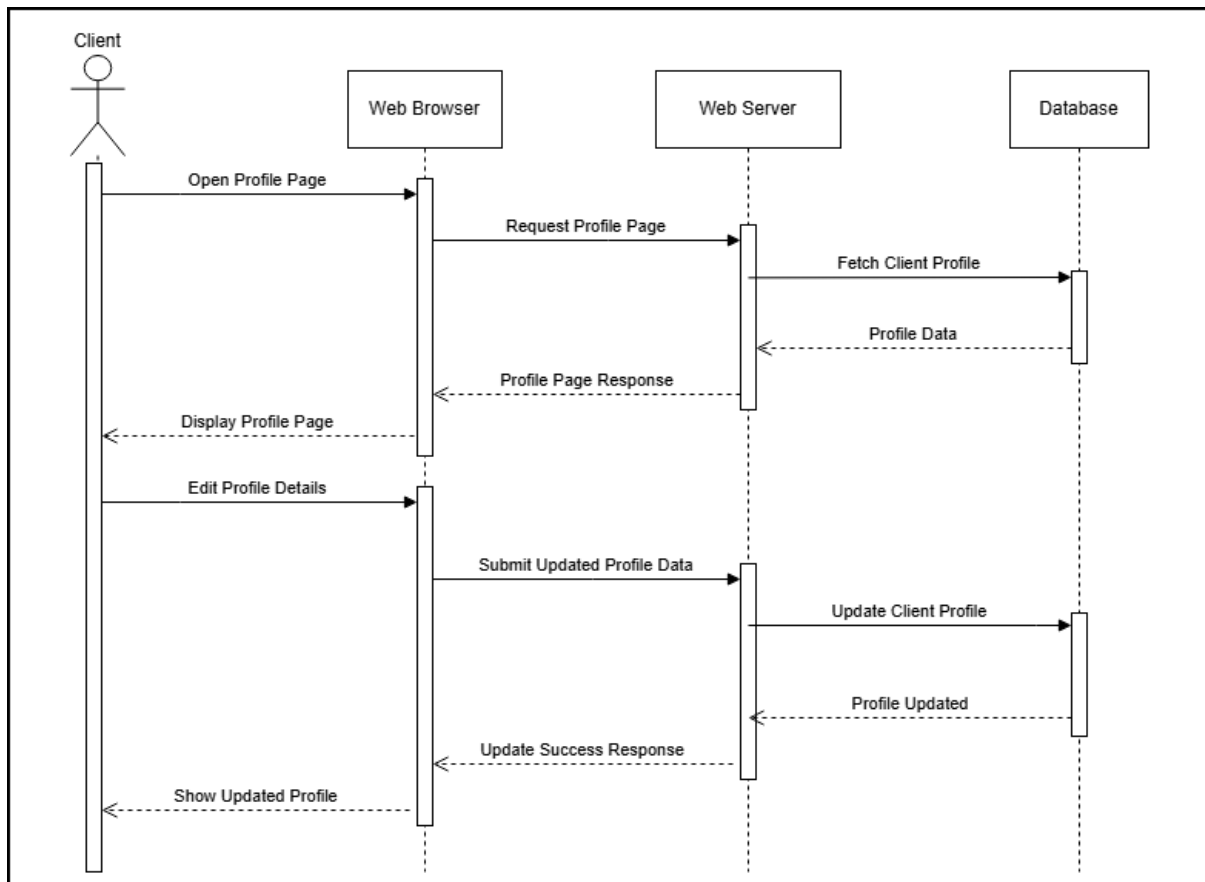


Figure 11: Client Update Profile Sequence Diagram

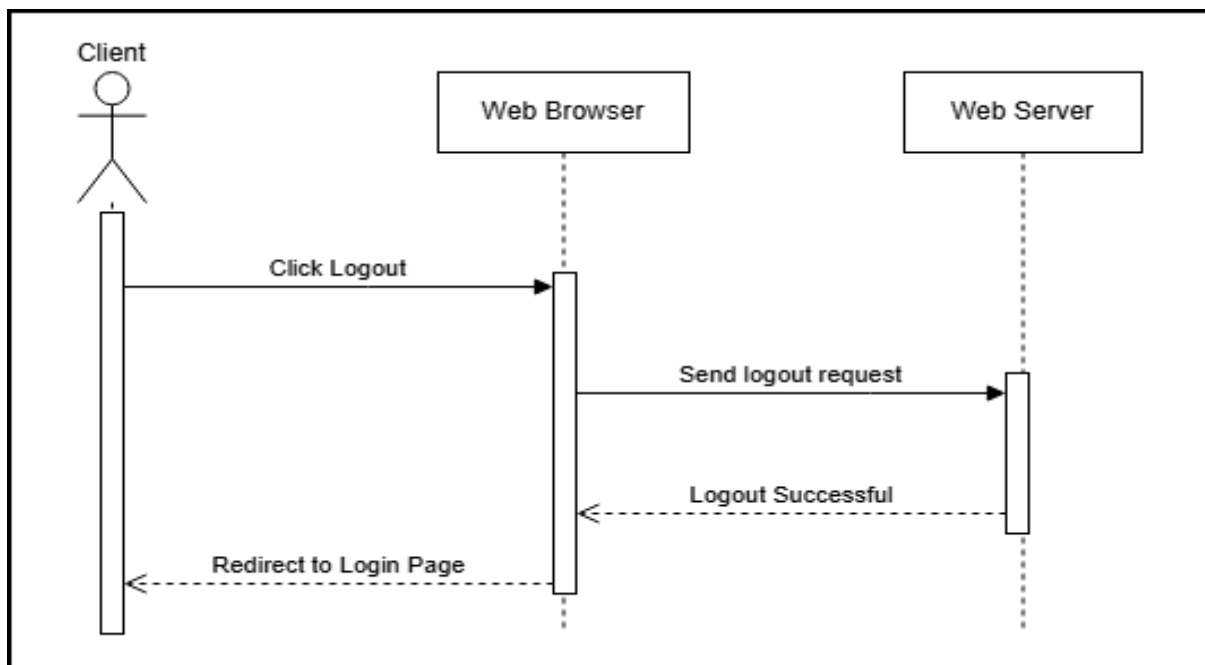


Figure 12: Client Logout Sequence Diagram

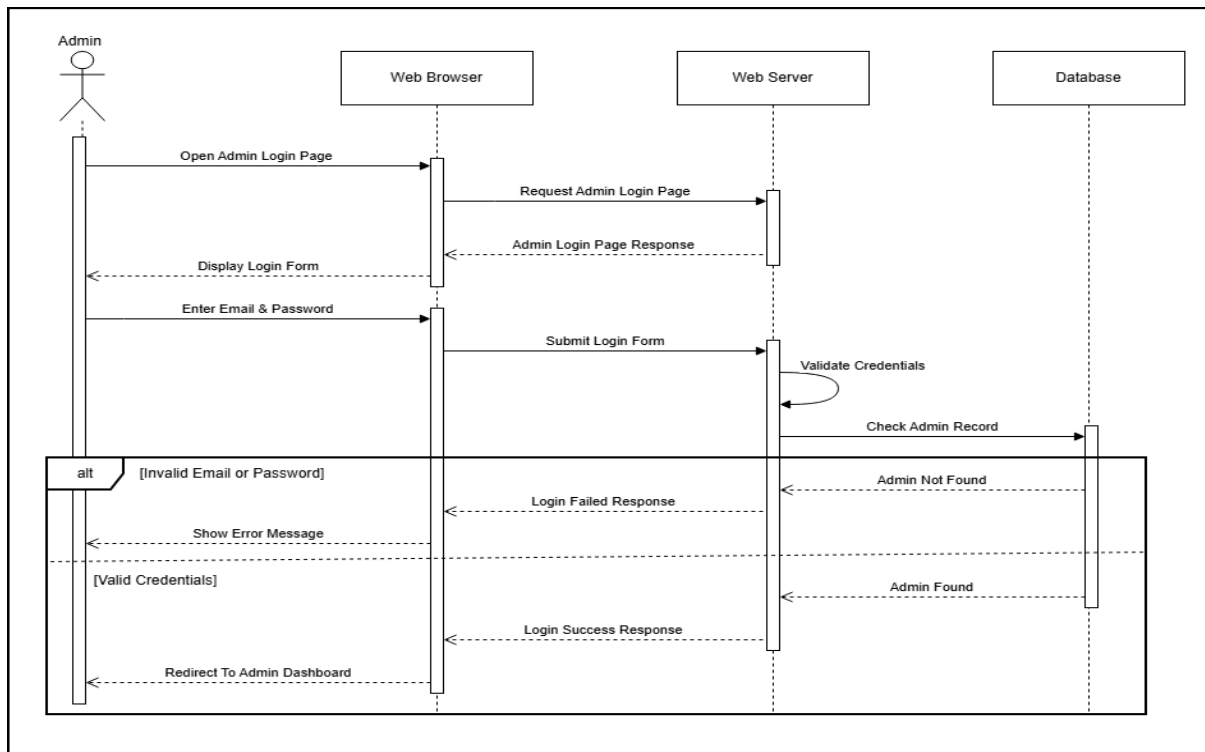


Figure 13: Admin Login Sequence Diagram

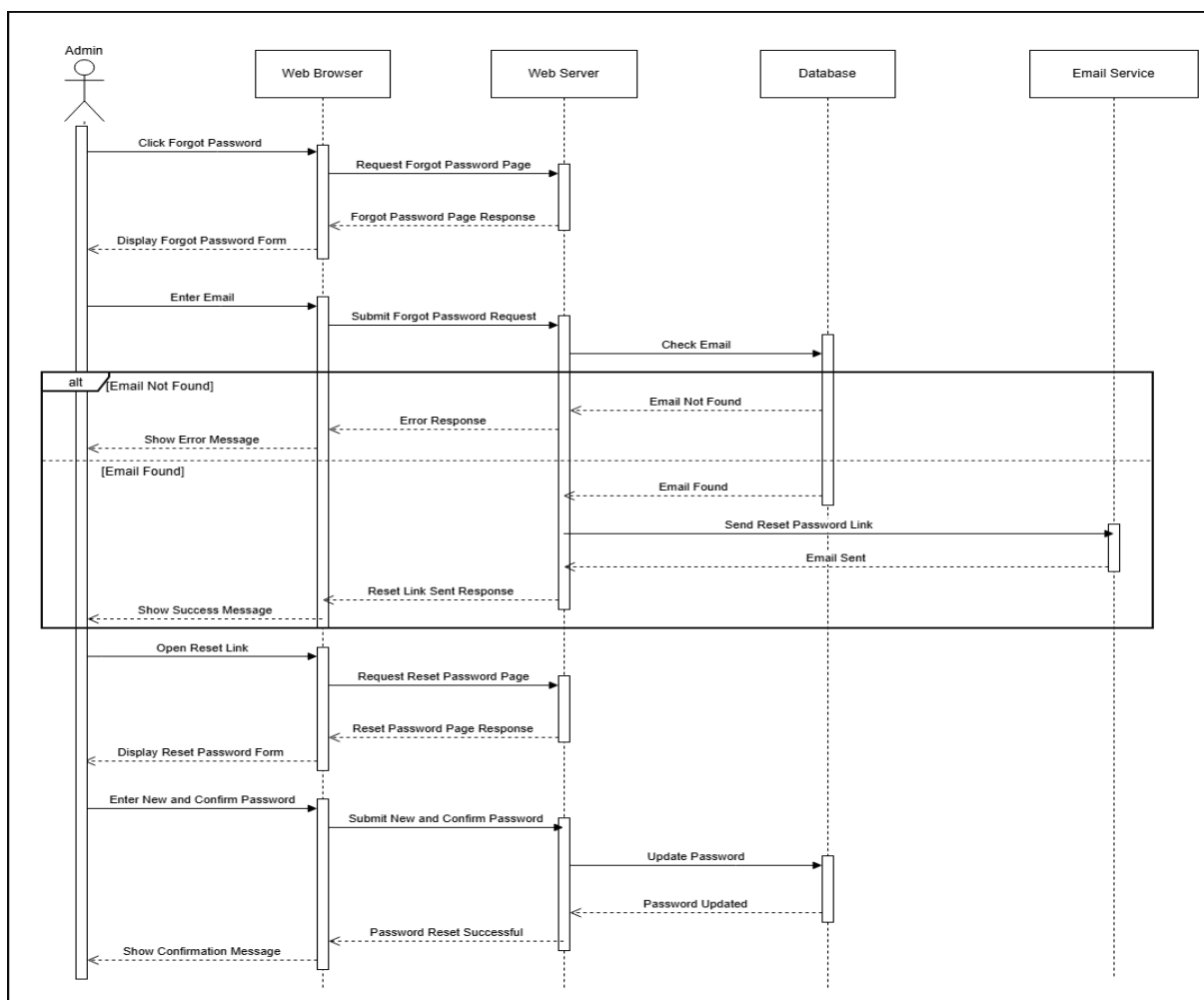


Figure 14: Admin Forgot Password Sequence Diagram

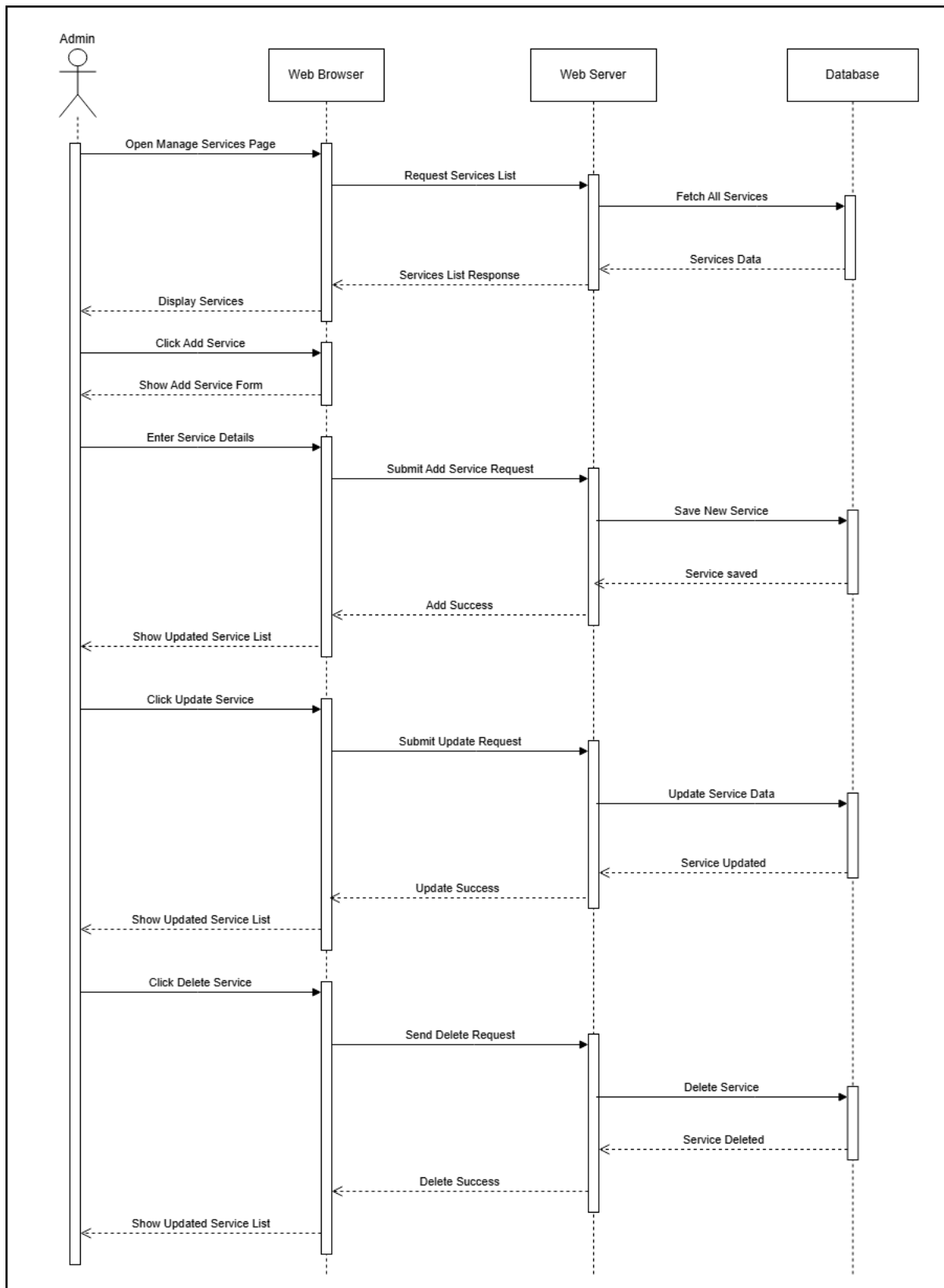


Figure 15: Manage Services Sequence Diagram

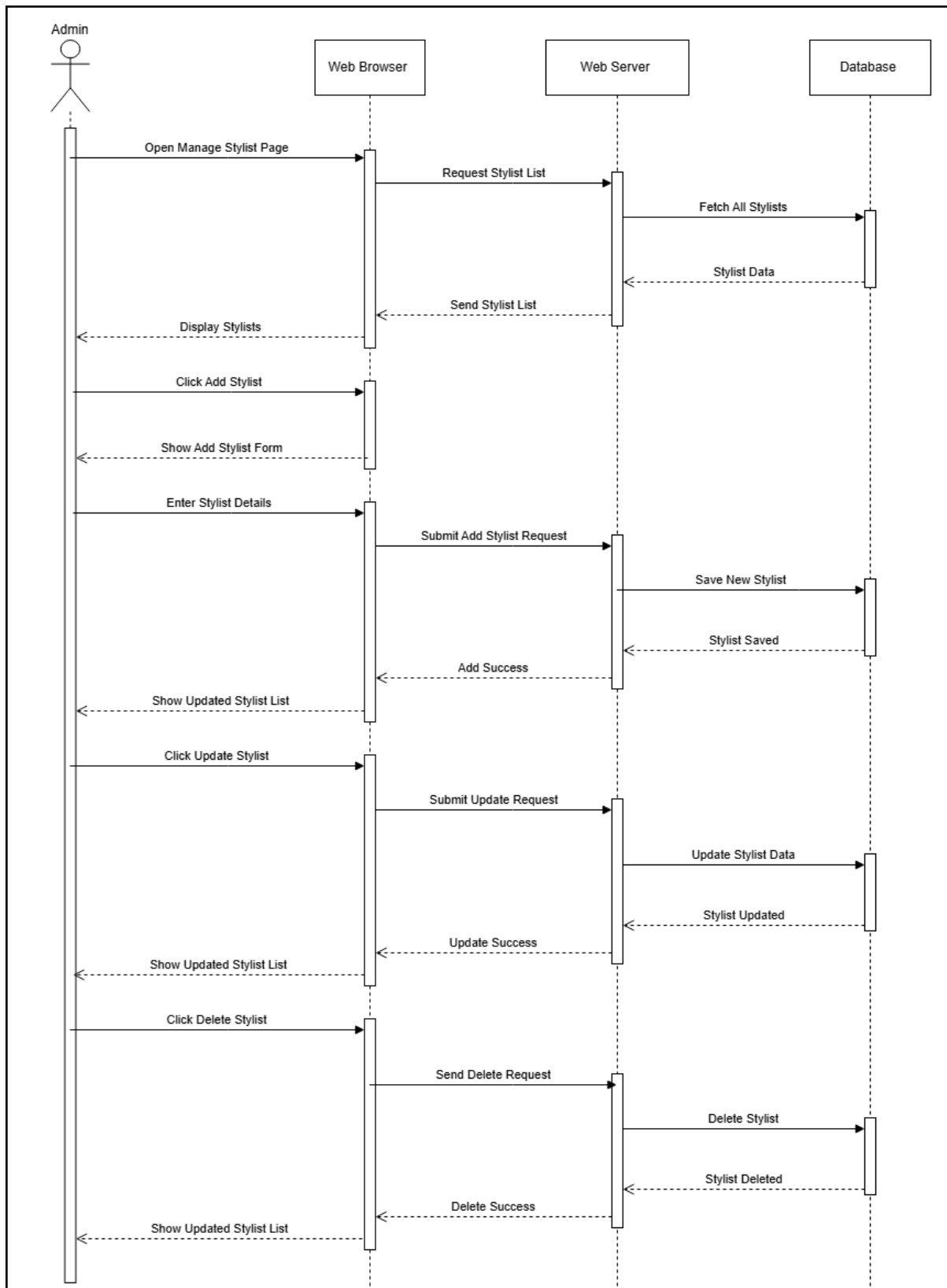


Figure 16: Manage Stylists Sequence Diagram

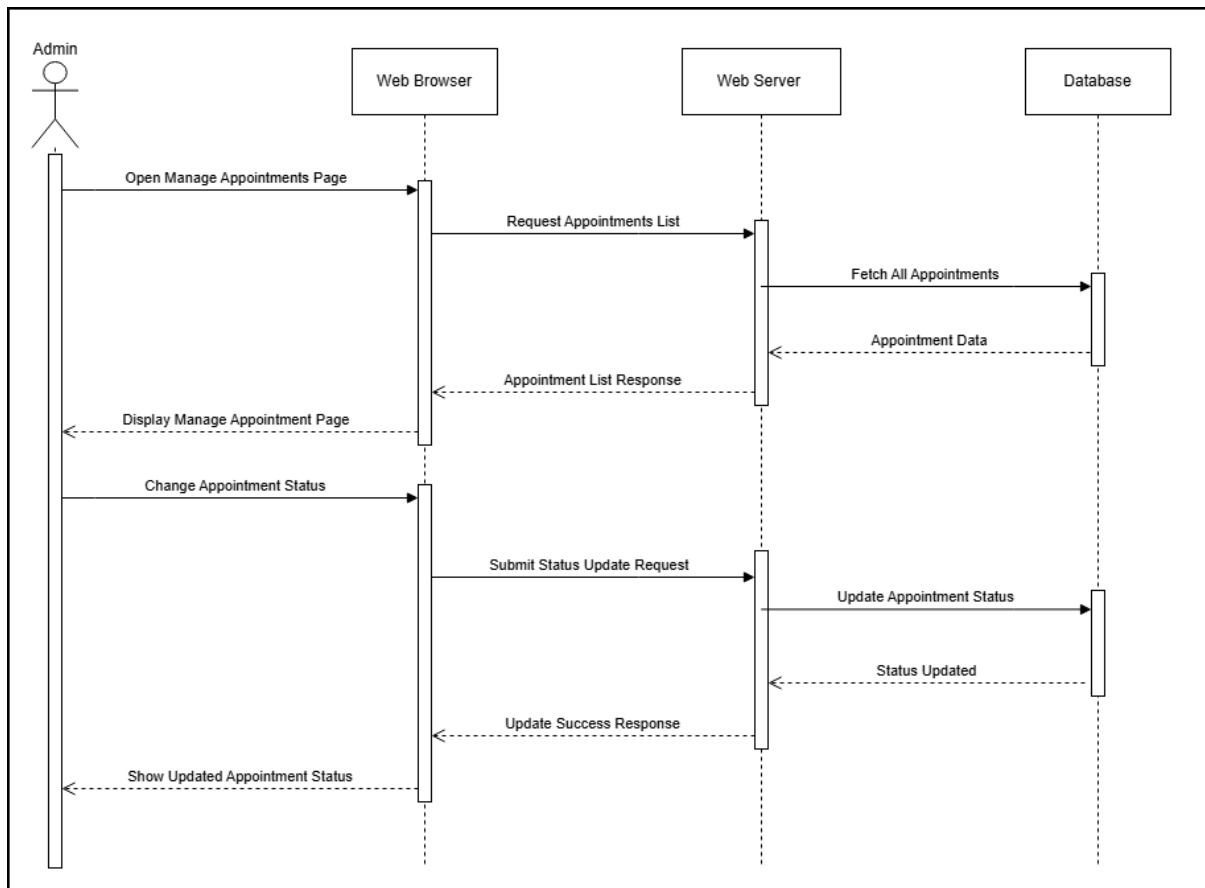


Figure 17: Manage Appointment Sequence Diagram

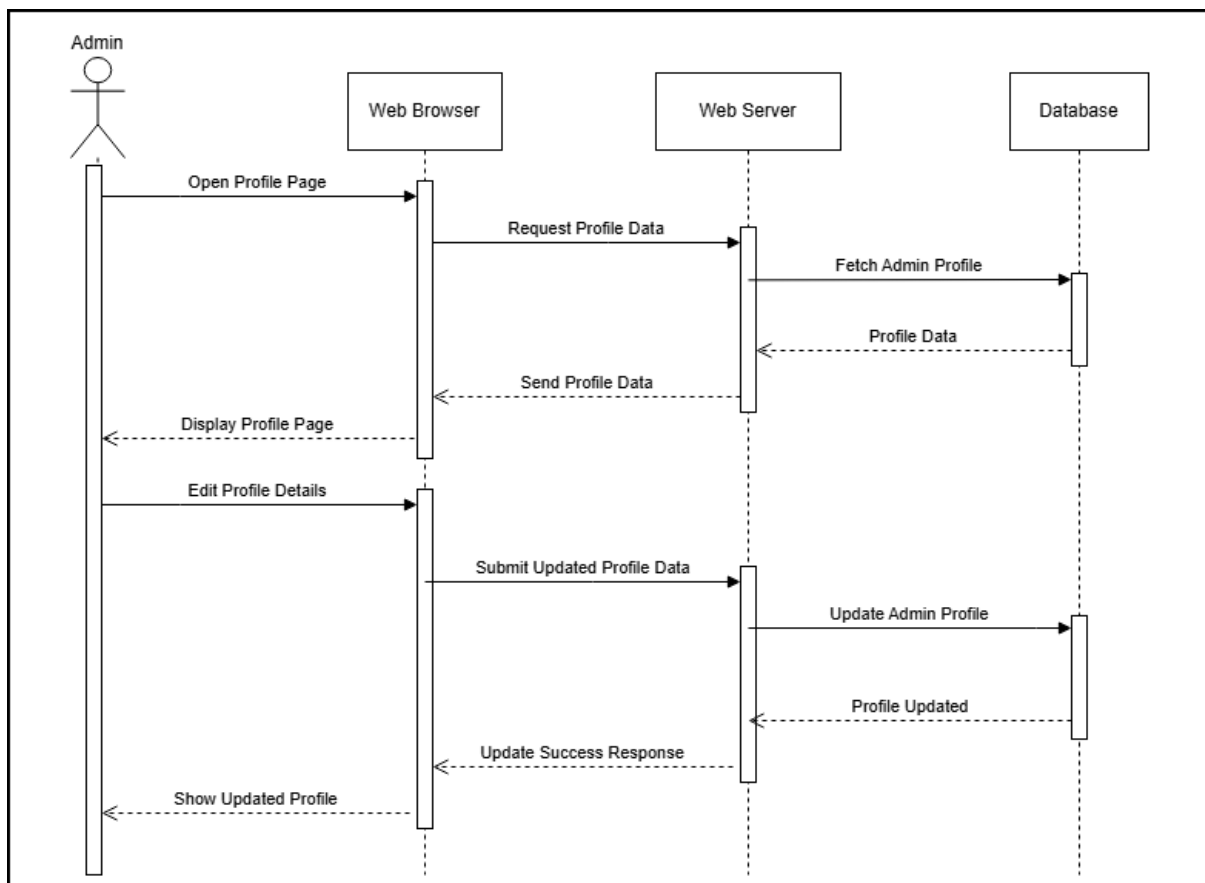


Figure 18: Admin Update Profile Sequence Diagram

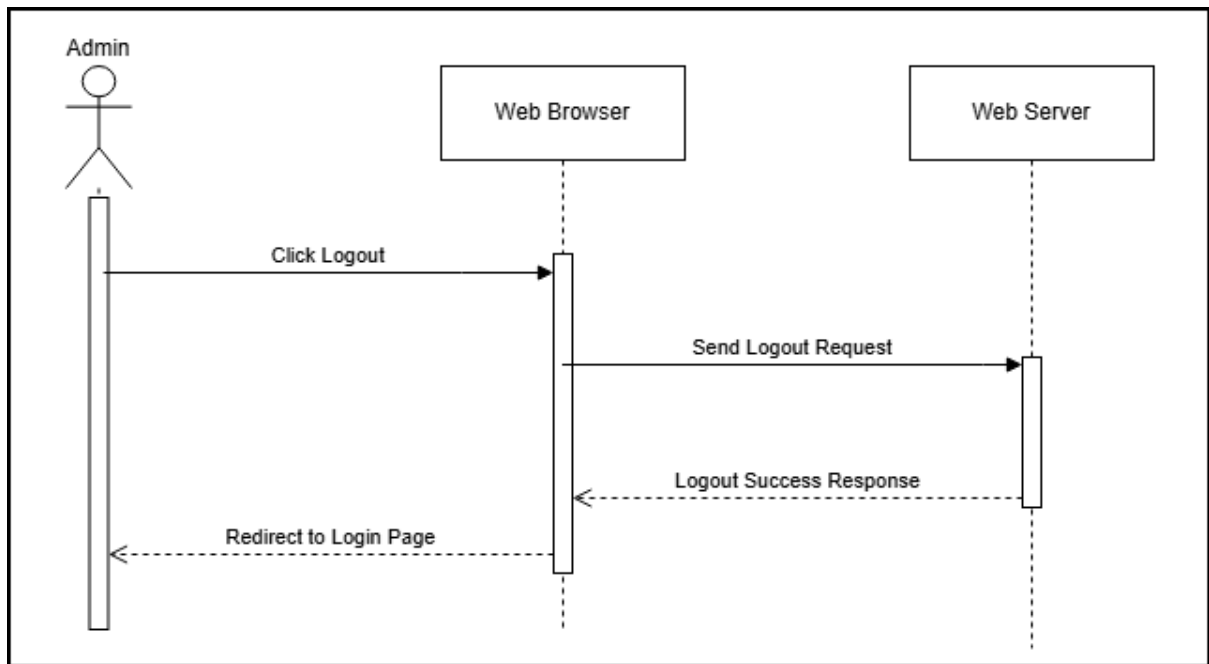


Figure 19: Admin Logout Sequence Diagram

3. State Transition Diagram

The state transition diagram will show how an appointment will change its state in the system. It will move through states liked booked, paid, cancelled and completed.

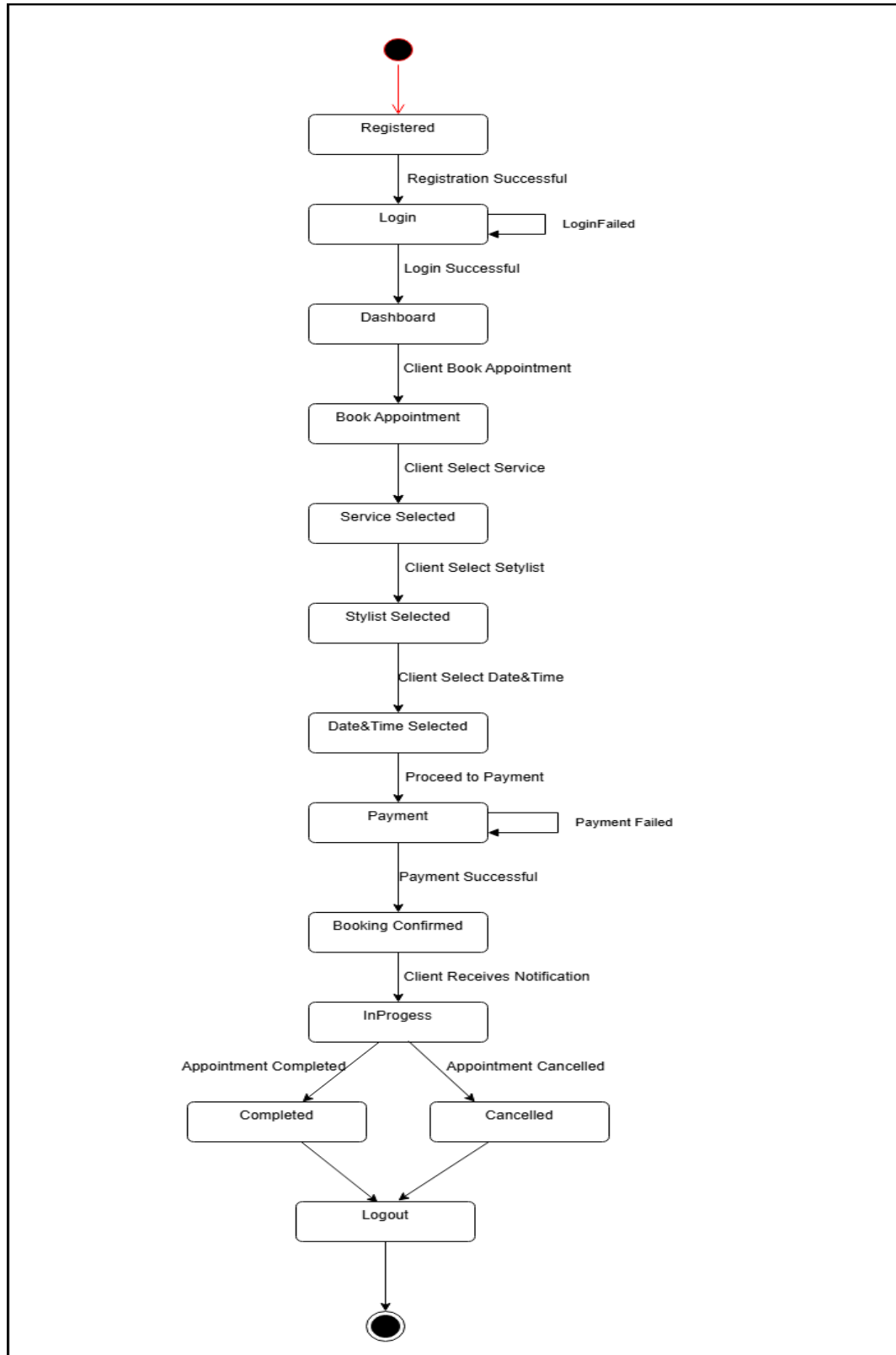


Figure 20: Client State Diagram

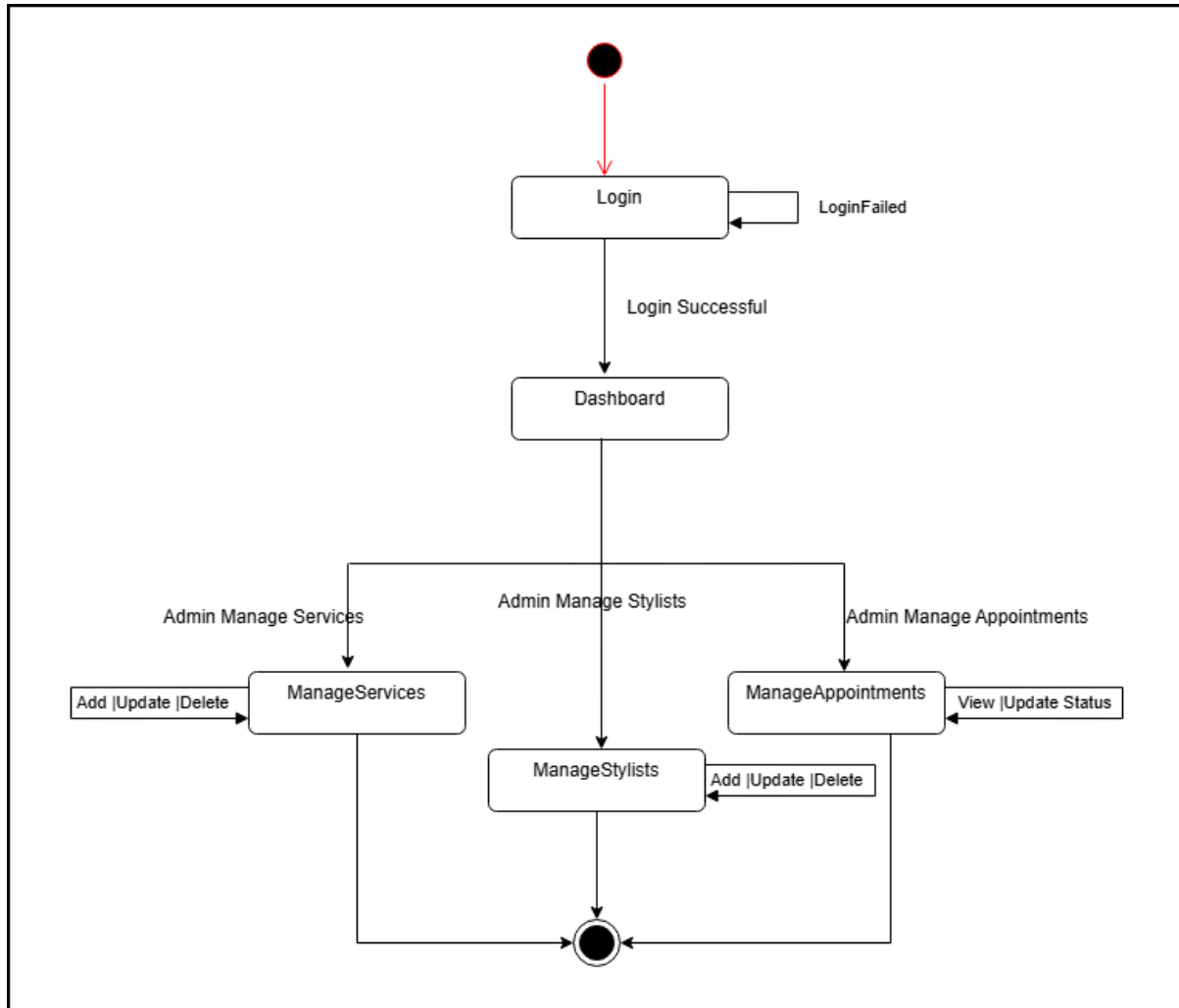


Figure 21: Admin State Diagram

4.5 Architectural Design

The Beauty Blush Salon Booking System will follow Client-Server, Layered and MVC architecture. These architectures will help the system work smoothly, remain organized and easy to manage.

- Client-Server architecture will allow clients and admin to use the system through a web interface, while all processing will be handled on the server.
- Layered architecture will divide the system into layers where each layer will have a specific responsibility.
- MVC architecture will separate data, user interface and control logic to keep the system clear and maintainable.

System Architecture Components:

The system will include a UML Component Diagram that will show the main components work together. It will represent frontend, backend, database and external services used in the Beauty Blush Salon system.

Frontend Layer (User Interface Components):

The frontend layer will handle all user screens and interactions.

- Client Registration UI
- Client Login UI
- Admin Login UI
- Client Dashboard UI
- Admin Dashboard UI
- Profile Management UI
- View Services UI
- Manage Services UI
- Stylist Selection UI
- Manage Stylists UI
- Payment UI
- Appointment Booking UI
- Manage Appointment UI
- Logout UI

Backend Layer (Logic / Controller Components):

The backend layer will handle system operations and business logic.

- Client Controller
- Authentication Controller
- Dashboard Controller
- Admin Controller
- Profile Controller
- Service Controller
- Stylist Controller
- Payment Controller
- Appointment Controller
- Notification Controller

- Logout Controller

Data Layer (Models / Database Components):

The Data Layer will store and manage all system data.

- Client Model
- Admin Model
- Service Model
- Stylist Model
- Appointment Model
- Payment Model
- Notification Model

External Services:

External services will support system functionality.

- Email Service
- Online Payment Gateway

Database:

- Main Database (MongoDB)

1. UML Component diagram

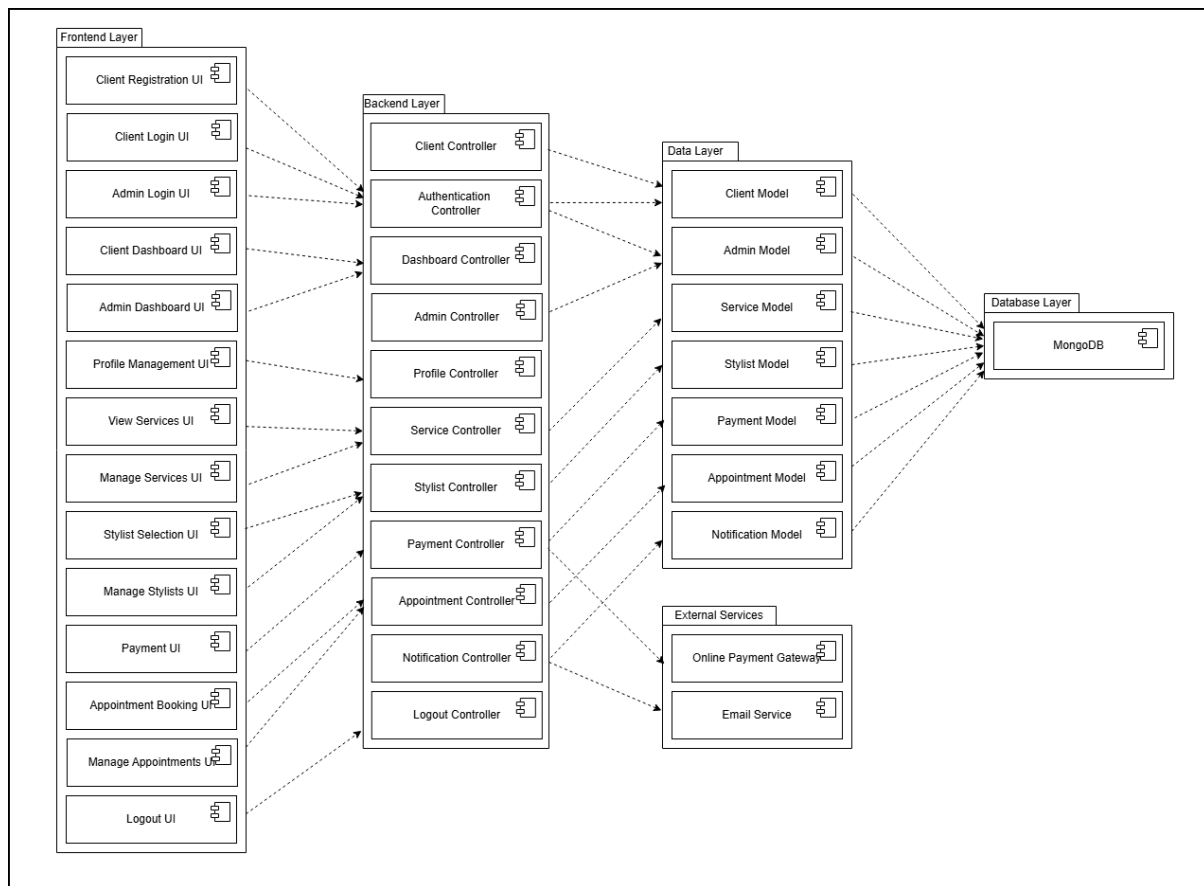


Figure 22: Component Diagram

4.6 User Interface Design

The user interface will provide a simple and easy way for any users to interact with the Beauty Blush Salon system. Users will be able to book appointments, make payments and view their booking details through clear and user-friendly screens.

1. Screen Images

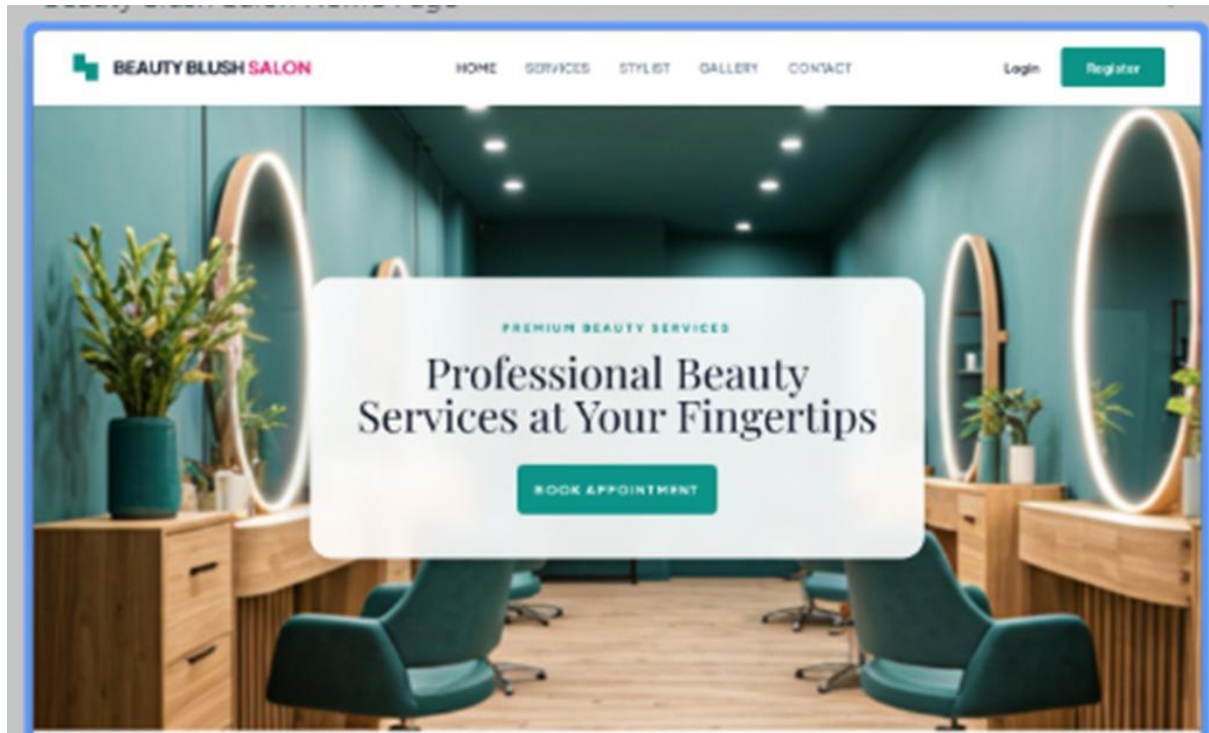


Figure 23: Home Page

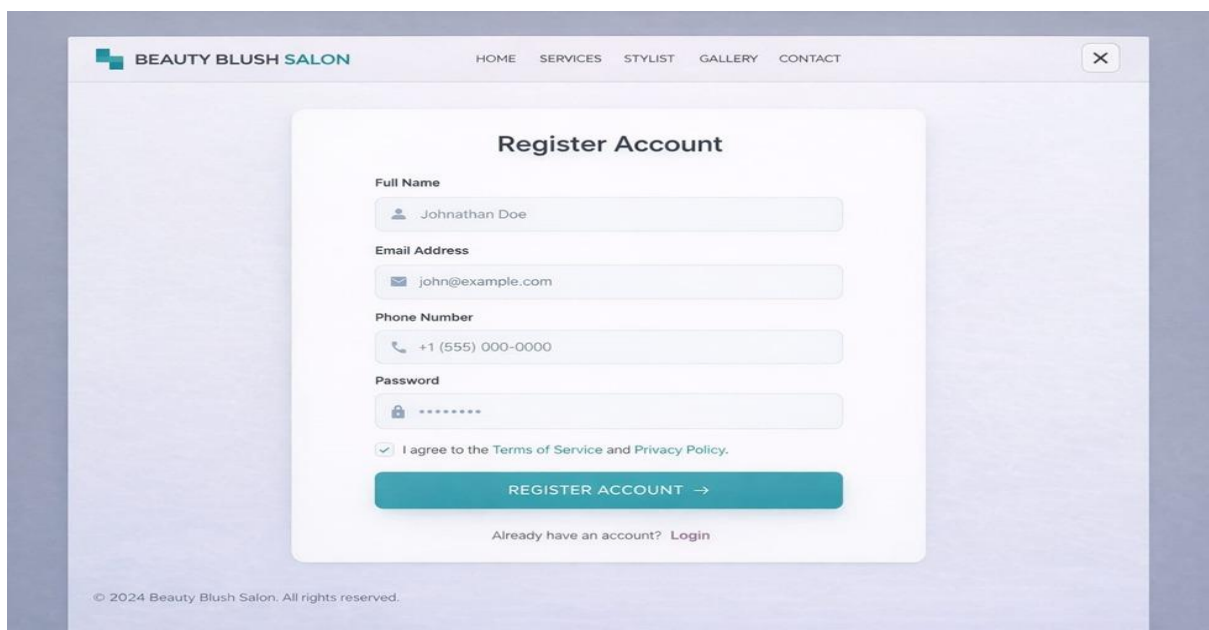


Figure 24: Registration Page

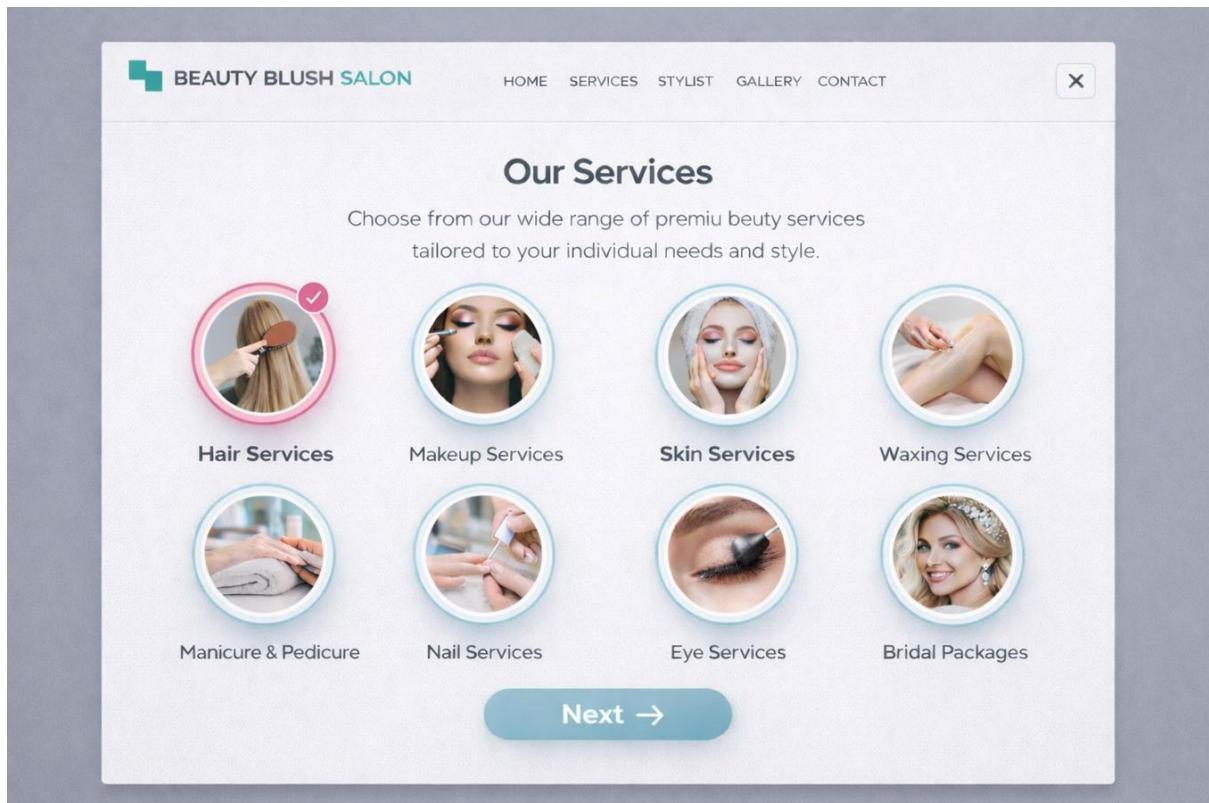


Figure 25: View Services

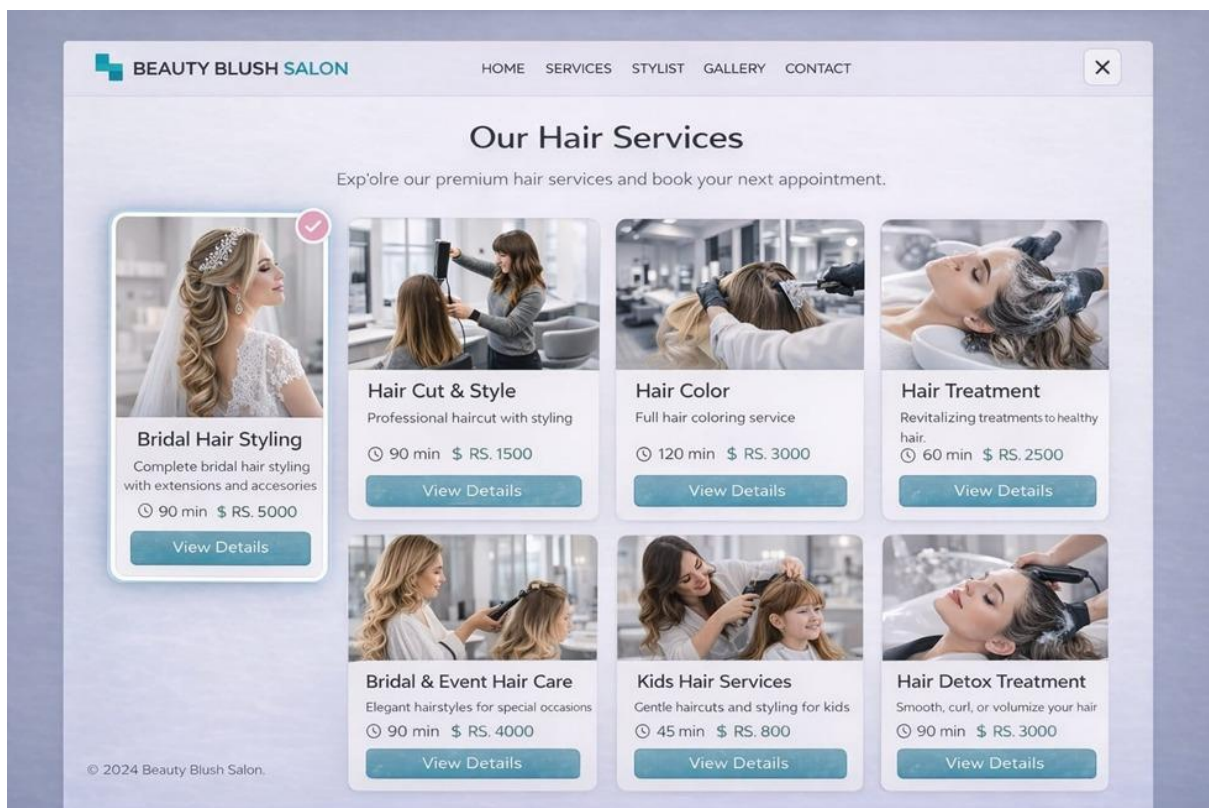


Figure 26: Select Service

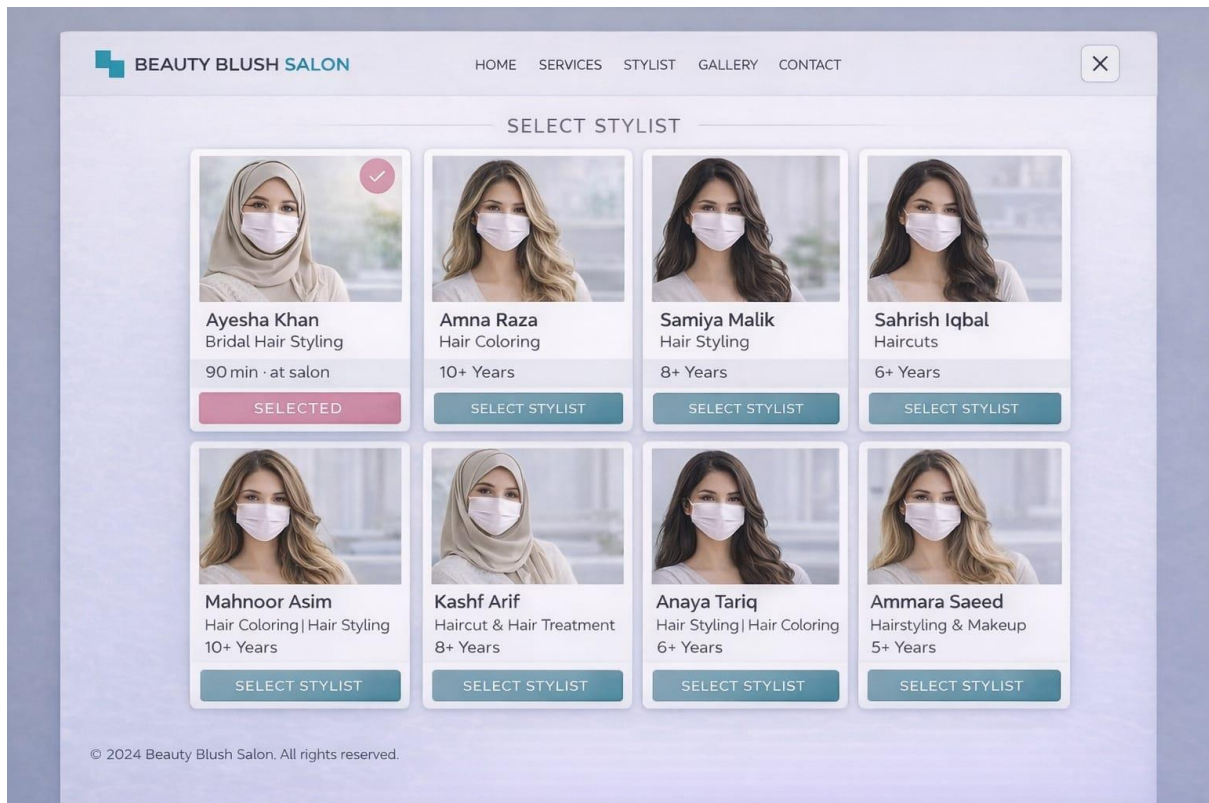


Figure 27: Select Stylist

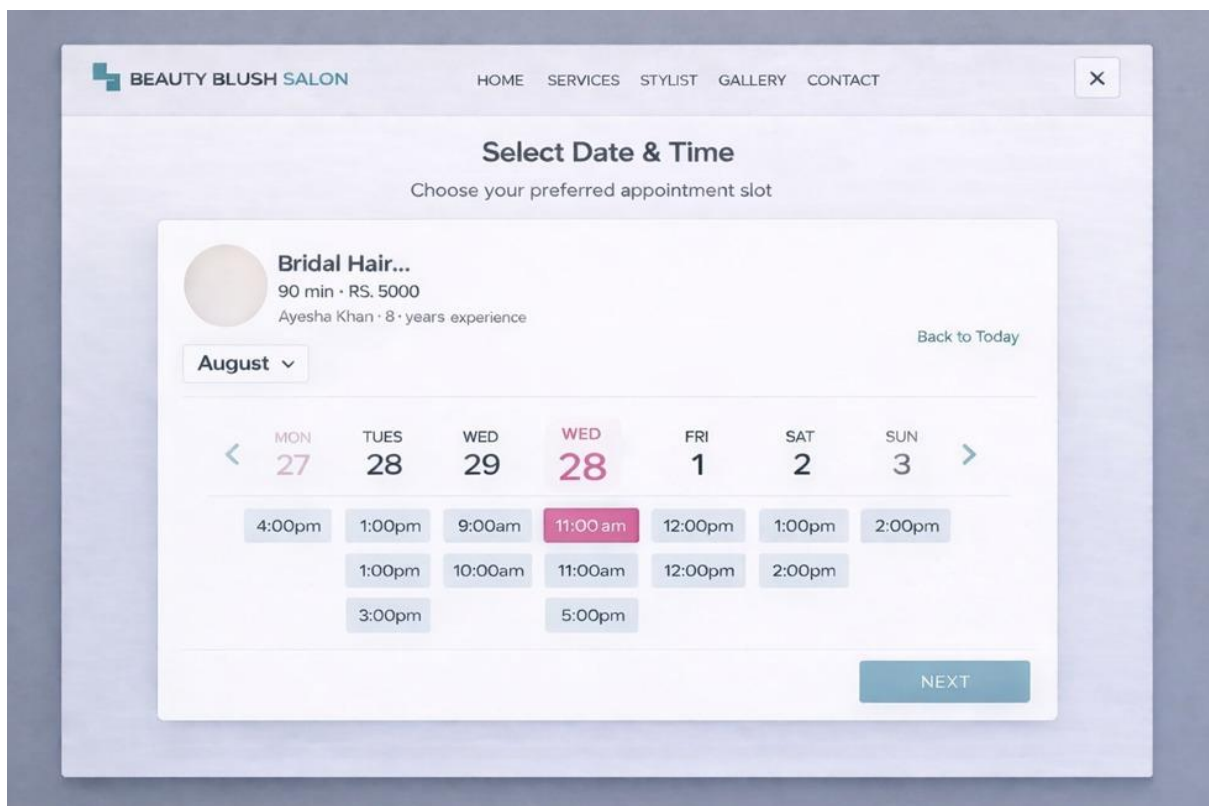


Figure 28: Select Date & Time

BEAUTY BLUSH SALON

HOME SERVICES STYLIST GALLERY CONTACT

X

Secure Payment

Complete your booking by making a partial payment for
Beauty Blush Salon

Enter Your Credit Card Details

Name On Card:

Card Number:

+ ✓

CVV

Expiration Date

Pay Now

SECURED BY STRIPE PAYMENT GATEWAY

Payment Summary

Amount Selected:

RS. 100.00

Total to Pay Now

RS. 100.00

What happens next?

✓ Your appointment will be confirmed immediately.

✓ You'll receive a confirmation email

✗ Remaining **RS. 4900** to be paid at salon

✓ Manage your appointment from the dashboard

Figure 29: Make Payment

BEAUTY BLUSH SALON

HOME SERVICES STYLIST GALLERY CONTACT

X

✓

Your Appointment
Has Been Booked Successfully

Your booking for **Bridal Hair Styling** has been successfully
successfully scheduled.

STYLIST



Ayesha Khan

✂

DATE

TIME

Wed, Aug 28, 2024

11:00 AM

✓ Payment Successful

RS. 100

🏠 Back to Home

Figure 30: Appointment Booked

4.7 Summary

This chapter described the complete system design of the beauty of the Beauty Blush Salon web-based booking system. It explained the system architecture, design models and user interface design used in the system. Class diagram, sequence diagram and state transition diagram were used to show system structure and behavior. The design focuses on simplicity, reliability and securely handling of appointments and payments to support smooth system implementation.

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